A Parametric Family of Primes via Quadratic Forms

Heuristic Distribution of the Minimal Offset, Large-Scale Numerical Validation, and Statistical Control of Spurious Spectral Correlations

Hassane BAKKAOUI
Independent Researcher
bakkahassa@hotmail.com

April 2026 — Revised Edition

Epistemic key (read first). This paper uses three levels of certainty, signalled throughout by coloured boxes:

Rigorous	Unconditional result, proved from classical theorems.
Heuristic	Conditional on Bateman–Horn (H1) and/or effective-independence model (H2′).
Negative	Rigorous invalidation of a previously reported claim.

Every claim not tagged with one of these labels is an explicitly labelled empirical observation or an open problem. No result is presented as an unconditional breakthrough.

Abstract

Context and objects of study. We study the parametric family of integers $p_{k,m,\varepsilon,q} = km(m+1) + \varepsilon + 2kq$ ($k, m \in \mathbb{N}^*$, $\varepsilon \in \{\pm 1\}$, $q \in \mathbb{Z}$), where $km(m+1)$ is a pronic multiple of k and q is the search offset. For fixed m, this is an arithmetic progression with step $2k$; the *minimal offset* $q_{\min}(k, m)$ is the smallest $|q|$ for which $p_{k,m,\varepsilon,q}$ is prime. The family generalises the classical $p \equiv \pm 1 \pmod 6$ observation and encompasses three geometrically structured cases: $k = 3$ (centered hexagonal lattice), $k = 15$ (cryptographic sieve), and $k = 21$ (main statistical study, 10^6 values).

Main heuristic law (conditional). Under the Bateman–Horn conjecture (H1) and the effective-independence model (H2′) — which asserts that primality correlations at distinct offsets vanish at rate $O(1/\ln m)$, consistently with Cramér–Granville [11, 12] — we derive the heuristic law

$$\mathbb{E}\big[|q_{\min}(k, m)|\big] \sim \frac{\ln m}{C_k},$$

where $C_k = C(f_{k,+}) + C(f_{k,-})$ is the aggregate Bateman–Horn constant (computed by Euler product over 17,984 primes, error $< 0.1\%$). A distributional refinement conjectures that $|q_{\min}| \cdot C_k / \ln m$ converges in distribution to Exp(1). Both results are heuristic: their derivation rests on H1 and H2′, neither of which is proved.

Empirical validation and correction factor. For $k = 21$: the Bateman–Horn aggregate $C_{21} = 8.206$ gives a theoretical slope $1/C_{21} = 0.122$. The empirical law $\mathbb{E}[|q_{\min}|] \approx 0.32 \ln m - 0.45$ ($R^2 = 0.984$, 10^6 values, bootstrap 95% CI for slope: $[0.31, 0.33]$) reveals an empirical correction factor $\rho_{21} = 0.32 \times C_{21} = 2.626 \pm 0.08$. This factor, heuristically decomposed into finite-m, bidirectional-search, and Cramér–Granville contributions, reflects the numerical fingerprint of the correction term $\varepsilon_k(m)$ of (H2′) at accessible scales. A chi-squared test on conditional bins ($m \in [10^4, 10^{4.5}]$,

15 equiprobable classes) yields $\chi^2 = 13.1$, df $= 14$, $p = 0.52$: the rescaled offset is compatible with the geometric distribution at all tested scales.

Robust structural decomposition. The cumulative functional $Q(p_N) = \sum_{n=1}^{N} c_n$, where $c_n = q_n^\circ + \varepsilon_n^\circ$, satisfies $Q \approx \alpha\sqrt{p} + \beta D_N$ with $R^2 > 0.967$, confirmed under three independent parameter conventions, where D_N is the cumulative deviation from Weyl equidistribution of $\{\sqrt{p/3}\}$. This decomposition accounts for 96.7% of the variance of Q and is independent of computational convention — it is the sole non-conditional structural result beyond Theorem 2.2.

Decisive negative result. Earlier spectral correlations between Q and non-trivial zeros of $\zeta(s)$ (Pearson p-values as small as 3.8×10^{-142}) are established as statistical artefacts of spurious regression between integrated ($I(1)$) processes [27], not genuine signals. Three independent permutation tests (shuffle, block, Theiler surrogate) yield z-scores in $[-0.23\sigma, -0.02\sigma]$, and a decisive test on 10^8 primes finds $R^2 = 1.16 \times 10^{-7}$ ($z = -2.20\sigma$), with no convergence over four decades of scale.

Epistemic status. All quantitative asymptotic results are heuristic and conditional on H1 and/or H2′. The sole unconditional result is Theorem 2.2 (Dirichlet's theorem). No result is presented as an unconditional breakthrough. This work is a contribution to experimental mathematics and computational number theory.

Keywords: prime numbers, quadratic forms, Bateman–Horn conjecture, Cramér–Granville model, heuristic laws, permutation tests, spurious regression, experimental mathematics.

MSC 2020: 11N32, 11Y11, 11M41, 94A60, 62P99.

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1 Introduction

1.1 Motivation and Context

Every prime $p > 3$ satisfies $p \equiv \pm 1 \pmod 6$, placing it within one step of the nearest multiple of 6. This elementary observation is the prototype of a *parametric prime family*: a structured doubly-indexed sequence $\{p_m\}_m$ in which each p_m is extracted from an arithmetic progression centred at a quadratic base, with a computable search cost.

The family studied here,

$$p_{k,m,\varepsilon,q} \;=\; km(m{+}1) + \varepsilon + 2kq, \tag{1}$$

generalises the $p \equiv \pm 1 \pmod 6$ observation by replacing 6 with $2k$, the linear variable with the pronic number $m(m{+}1)$, and making the search offset q explicit. For fixed m, equation (1) defines an arithmetic progression with common difference $2k$, to which Dirichlet's theorem applies unconditionally [1, 2].

1.2 Results at Three Levels of Rigor

The paper operates at three distinct epistemic levels, collected here for immediate orientation.

Level 1 — Rigorous and unconditional.

(R1) (*Theorem 2.2, Dirichlet 1837.*) For squarefree k and any $m \ge 1$, $\varepsilon \in \{\pm 1\}$: the family $\{km(m{+}1) + \varepsilon + 2kq : q \in \mathbb{Z}\}$ contains infinitely many primes. In particular, $q_{\min}(k,m,\varepsilon)$ is always well-defined.

(R2) (*Theorem D.2, under GRH.*) Assume the Generalised Riemann Hypothesis for all Dirichlet L-functions. Then for squarefree k and all sufficiently large m:

$$|q_{\min}(k,m,\varepsilon)| \;=\; O_k(m \log m).$$

This is the best currently available upper bound on $q_{\min}$; the gap to the heuristic $O(\ln m)$ is the central open problem (O1–O2 of Section 9).

Level 2 — Heuristic, conditional on H1+H2′.

(H) (*Theorem 3.2 and Conjecture 3.3.*) Under the Bateman–Horn conjecture (H1) and the effective-independence model (H2′):

$$\mathbb{E}\big[|q_{\min}(k,m)|\big] \;\sim\; \frac{\ln m}{C_k} \qquad \text{as } m \to \infty,$$

and the rescaled variable $|q_{\min}| \cdot C_k/\ln m$ is conjectured to converge in distribution to Exp(1). These are *plausibility arguments* conditional on two unproven hypotheses; the derivation is not a proof (Section 3).

Level 3 — Empirical observations (illustrative, not demonstrative).

For $k = 21$, $m \in [10^3, 10^6]$: the empirical law $\mathbb{E}[|q_{\min}|] \approx 0.32 \ln m - 0.45$ ($R^2 = 0.984$, 95% CI for slope: $[0.31, 0.33]$) is *consistent with* the heuristic law (H) up to a finite-scale correction factor $\rho_{21} = 2.626$. These numerical results are illustrative of the behaviour predicted by (H); they do not constitute a proof of the asymptotic law and should not be interpreted as such.

The main heuristic law (H) is the principal new contribution of this paper. Result (R2) is recalled from the literature (it follows from a classical argument under GRH; see Appendix D and [2]) and is reproduced here to anchor the heuristic claim between two rigorous bounds.

1.3 Relation to Existing Literature and Claims of Novelty

Relation to Dirichlet and arithmetic progressions. The infinitude of primes in (1) (Result (R1)) is an immediate consequence of Dirichlet's theorem [1, 2]. It is stated only for completeness. The upper bound (R2) under GRH is a standard consequence of the effective Chebotarev density theorem [2]; no originality is claimed for it. Both are recalled to provide the rigorous envelope within which the heuristic claims (H) must eventually be situated.

Relation to Bateman–Horn and probabilistic models. The heuristic law (H) is an application of the Bateman–Horn conjecture [5, 6] to the specific polynomial family $f_{k,\varepsilon}(m) = km(m+1) + \varepsilon$, combined with the Cramér–Granville independence model [11, 12]. The Bateman–Horn conjecture, Cramér's model, and Weyl equidistribution [18] are classical; we apply them, not prove them. The principal novelty is the explicit identification of the family (1), the derivation of C_k as the quantitative search-cost parameter, and the systematic large-scale numerical study at scale 10^6–10^8.

What is genuinely new.

1. The heuristic law $\mathbb{E}[|q_{\min}|] \sim \ln m / C_k$, its derivation under H1+H2′, and its systematic comparison with data across three values of k at scale 10^6. The comparison is *consistent with* the law; it does not constitute a proof.
2. Conjecture 3.3: the distributional prediction $|q_{\min}| \cdot C_k / \ln m \xrightarrow{d} \mathrm{Exp}(1)$, supported by a KS test and a chi-squared test (Section 5.4).
3. The structural decomposition $Q \approx \alpha\sqrt{p} + \beta D_N$ ($R^2 > 0.967$), reproducible under three independent parameter conventions (Section 6).
4. The application of three independent permutation tests (T1, T2, T3) to identify spectral correlations in Q as spurious-regression artefacts (Section 7), and the decisive validation on 10^8 primes (Section 8).

No result in items 1–4 is presented as unconditional; each is explicitly labelled according to its epistemic status.

1.4 Organisation

Sections 2–4 establish the theoretical framework and Bateman–Horn constants. Section 5 presents large-scale numerical validation ($k = 21$, 10^6 values). Section 6 establishes the $Q \approx \alpha\sqrt{p} + \beta D_N$ decomposition. Sections 7 and 8 constitute the core statistical methodology and decisive negative result. Section 9 discusses limitations and open problems.

Secondary material appears in appendices: conjectures C1–C2–C3 for $k = 3$ (Appendix A), the cryptographic application for $k = 15$ (Appendix B), spectral analogies retained for documentary purposes (Appendix C), and the rigorous bounds on $q_{\min}$ including the GRH-

conditional upper bound (Appendix D). Section 6.3 documents the statistical near-independence of the geometric factor m_r from the arithmetic offset $(q_r+\varepsilon_r)$; this is an empirical observation, not a theorem.

2 Theoretical Foundations

2.1 The Parametric Family

Definition 2.1 (Parametric form and minimal offset)**.** For $(k,m) \in (\mathbb{N}^*)^2$, $\varepsilon \in \{\pm 1\}$, $q \in \mathbb{Z}$, define $p_{k,m,\varepsilon,q} = km(m+1) + \varepsilon + 2kq$. The *minimal offset* is $q_{\min}(k,m,\varepsilon) = \operatorname{argmin}_{q\in\mathbb{Z}}\{|q| : p_{k,m,\varepsilon,q} \text{ prime}\}$.

For fixed m, (1) is the arithmetic progression $A + 2k \cdot q$ with $A = km(m+1) + \varepsilon$ and step $d = 2k$. Table 1 summarises the three cases studied.

Table 1: Three cases studied: structural parameters and Bateman–Horn constants.

k	$2k$	$\varphi(2k)$	C_k	Geometric structure
3	6	2	6.145	Centered hexagonal numbers ($\mathbb{Q}(\sqrt{-3})$) [20]
15	30	8	6.814	30-step sieve (cryptographic application)
21	42	12	8.206	Main statistical study (10^6 values)

2.2 Unconditional Infinitude

Theorem 2.2 (Infinitude of primes — unconditional)**.** *Let $k \in \mathbb{N}^*$ be squarefree and $m \geq 1$. For each $\varepsilon \in \{\pm 1\}$, the family $\{km(m+1) + \varepsilon + 2kq : q \in \mathbb{Z}\}$ contains infinitely many primes.*

Proof. Since k is squarefree: (a) $m(m+1)$ is even, so $km(m+1) \equiv 0 \pmod 2$, giving $A = km(m+1) + \varepsilon \equiv \varepsilon \equiv \pm 1 \pmod 2$, hence $\gcd(A, 2) = 1$. (b) For each odd prime $p \mid k$: $km(m+1) \equiv 0 \pmod p$, so $A \equiv \varepsilon \not\equiv 0 \pmod p$. Therefore $\gcd(A, 2k) = 1$, and Dirichlet's theorem (1837) applies [1, 2]. □

Remark 2.3 (Rigorous results available in this paper)**.** Theorem 2.2 is the sole *unconditional* result. It guarantees that $q_{\min}$ is always well-defined but gives no bound on its size. Under the Generalised Riemann Hypothesis, Theorem D.2 (Appendix D) provides the non-trivial bound $|q_{\min}| = O_k(m \log m)$, which is the best rigorously available. All other quantitative results in Sections 3–8 are heuristic (conditional on H1+H2′) or empirical. The heuristic $O(\ln m)$ prediction is the primary numerical target of this paper, but it is not proved.

Corollary 2.4 (Density lower bound)**.** *Under the conditions of Theorem 2.2: $\#\{|q| \leq Q : p_{k,m,\varepsilon,q} \text{ prime}\} = \Omega(Q/\ln(kQ^2))$ as $Q \to \infty$.*

2.3 Basic Structural Properties

Proposition 2.5 (Search range)**.** *$p_{k,m,\varepsilon,-m} = km(m-1) + \varepsilon = p_{k,m-1,\varepsilon,0}$ and $p_{k,m-1,\varepsilon,m} = p_{k,m,\varepsilon,0}$. It suffices to search $q \in [-m, m]$; hence the trivial bound $|q_{\min}| \leq m$ (Theorem D.1).*

Lemma 2.6 (Non-divisibility for $k = 21$)**.** *For $k = 21 = 3 \times 7$, every element $p_{k,m,\varepsilon,q}$ is coprime to both 3 and 7.*

Remark 2.7 (Algebraic origin of the log-scale correlation)**.** For large m, the parametric form (1) yields, by direct Taylor expansion:

$$\ln p_{k,m,\varepsilon,q} \;=\; 2\ln m + \ln k + \frac{1}{m} + O\Big(\frac{q}{m^2}\Big).$$

For $k = 3$ this gives the asymptotic identity $\ln p \approx 2\ln m + \ln 3 \approx 2\ln m + 1.099$.

Consequence. Any empirical correlation $\rho(\ln p, \ln m)$ computed over the family is quasi-perfect (≈ 0.9997) *by algebraic necessity*, not by arithmetic coincidence. The theoretically imposed slope is $\alpha = 2$ and the intercept is $\beta_{\text{th}} = \ln 3 \approx 1.099$.

Informative content. The numerical interest lies not in the near-perfect correlation itself (which is trivially implied by Definition 1), but in the precise gap

$$\beta_{\text{emp}} - \ln k \;\approx\; \mathbb{E}\big[\ln\big(1 + \tfrac{1}{m}\big)\big] + \delta_{\text{disc}},$$

where $\delta_{\text{disc}} \approx 0.150$ for $k = 3$ encodes an arithmetic correction arising from the primality constraint and the non-uniform distribution of q for small m. Quantifying δ_{disc} rigorously is Open Problem O7.

3 The Main Heuristic Law

3.1 Derivation Under H1 and H2′

Standing assumptions for this section. (H1) The Bateman–Horn conjecture [5, 6] holds for the polynomials $f_{k,\varepsilon}(m) = km(m+1) + \varepsilon$ [8]. (H2′) Primality events at distinct offsets q are *effectively independent*: residual primality correlations are controlled by a term $\varepsilon_k(m) = O(1/\ln m)$ (Cramér–Granville model [11, 12, 13]). Both (H1) and (H2′) are *unproven*. Results labelled "heuristic" in this section are conditional on their truth.

Remark 3.1 (On the status of (H2′))**.** Strict independence of primality events at distinct offsets q is not expected to hold: Cramér's original model [11] already predicts correlations at scale $\ln m$, and Granville [12] establishes that true prime gaps are wider than the Cramér prediction by a factor $2e^{-\gamma} \approx 1.123$ asymptotically. Hypothesis (H2′) does *not* claim these correlations vanish; it asserts only that their net effect on the joint primality probability of candidates at a given step q is controlled by a multiplicative term $(1 + \varepsilon_k(m))$ with $\varepsilon_k(m) = O(1/\ln m)$ as $m \to \infty$ — precisely the form used in all standard Cramér-type heuristics; see Tao [13] for a careful formulation. The correction factor ρ_k of Section 3.3 is the numerical fingerprint of $\varepsilon_k(m)$ at the scales ($m \le 10^6$) accessible in this study.

Under (H1), the probability that the candidate at offset q is prime satisfies:

$$\Pr(p_{k,m,\varepsilon,q} \text{ prime}) \;\approx\; \frac{C(f_{k,\varepsilon})}{2\ln m}. \tag{2}$$

At each step $q \ge 1$, four candidates are tested ($\varepsilon = \pm 1$, q and $-q$); under (H2′) these events are effectively independent, giving a combined success probability per step:

$$p_q \;\approx\; \frac{C(f_{k,+}) + C(f_{k,-})}{2\ln m} \;=\; \frac{C_k}{2\ln m}. \tag{3}$$

Under (H2′), $|q_{\min}|$ is approximately *geometrically distributed* with parameter $p_q \approx C_k/(2\ln m)$, up to a correction term $\varepsilon_k(m) = O(1/\ln m)$ arising from residual primality correlations not fully absorbed by the Bateman–Horn constant [24, 12]. At finite m, the dominant contribution to $\varepsilon_k(m)$ is the Cramér–Granville factor, explaining the empirical correction ρ_k of Section 3.3.

Theorem 3.2 (Heuristic law for $q_{\min}$)**.** *Under* (H1) *and* (H2′)*, as $m \to \infty$:*

$$\mathbb{E}\big[|q_{\min}(k,m)|\big] \;\sim\; \frac{\ln m}{C_k}. \tag{4}$$

This is a heuristic prediction*, not a theorem. Its derivation is presented as a plausibility*

argument, not a proof.

3.2 Distributional Refinement

The geometric distribution model yields a more precise prediction than the mean alone.

Conjecture 3.3 (Exponential distributional limit)**.** *Under* (H1) *and* (H2′)*, as* $m \to \infty$*:*

$$\frac{C_k \cdot |q_{\min}(k,m)|}{\ln m} \xrightarrow{d} \mathrm{Exp}(1), \tag{5}$$

i.e., $\Pr\left(|q_{\min}| > t \cdot \frac{\ln m}{C_k}\right) \to e^{-t}$ *for all* $t > 0$*. This is a conjecture: the convergence in distribution is not proved; it requires controlling the rate of geometric convergence of* $|q_{\min}| \cdot C_k / \ln m$ *under H1+H2′ (open sub-problem of O5).*

This distributional conjecture is more informative than (4): it predicts the full tail behaviour, not only the mean. After empirical rescaling by ρ_{21} (see Section 3.3), a KS test on 10^6 values for $k = 21$ gives $D_{\mathrm{KS}} = 0.031$ with $p \approx 0.12$ (after subsampling for approximate independence), consistent with the exponential model at the 5% level. The conjecture is numerically supported but remains unproved; see Section 5.4 for the complete distributional validation, including a chi-squared test.

3.3 The Correction Factor ρ_k

The empirical slope (for $k = 21$) is 0.32, versus the theoretical prediction $1/C_{21} = 0.122$, giving $\rho_{21} = 0.32 \times C_{21} = 2.626$. Three effects contribute heuristically:

1. *Finite-m correction* (≈ 1.25): at finite m, the effective prime density in the progression is lower than the asymptotic BH value.
2. *Bidirectional search structure* (≈ 1.0): the $q = 0$ step has half the weight of steps $q \geq 1$.
3. *Cramér–Granville correction* (≈ 2.0): empirically, prime gaps near $km(m+1)$ are wider than the Cramér model [11] by a factor $2e^{-\gamma} \approx 1.123$ per direction [12], which compounds.

Total heuristic prediction: $\rho_{21} \approx 2.50$, consistent with 2.626.

Remark 3.4. The derivation of ρ_k as an explicit function of k is an open problem. Table 15 (Appendix B) shows that ρ_k varies between 0.40 and 1.45 across 27 values of k, with median ≈ 1. No closed-form formula is available.

3.4 Quantitative Structure of ρ_k: Empirical Analysis

The 25 empirical values of ρ_k listed in Table 15 permit a systematic investigation of what drives the correction factor. Two natural predictors are the Bateman–Horn aggregate C_k and the multiplicative complexity of k, measured by $\omega(k)$ (number of distinct prime factors).

Correlations. Computing Pearson correlations over the 25 data points:

$$r(\rho_k, C_k) = +0.85, \qquad r(\rho_k, \omega(k)) = -0.47.$$

The positive correlation with C_k ($r = 0.85$, $n = 25$) is consistent with higher-density families exhibiting stronger finite-scale discrepancies from the leading-order BH prediction, but $n = 25$ is too small to draw causal conclusions (see Remark 3.5). The negative correlation with $\omega(k)$ ($r = -0.47$) is a tentative pattern that may reflect sieve effects at finite m; it should not be over-interpreted.

Notable cases. For $k = 21 = 3 \times 7$ ($\omega = 2$): $\rho_{21} = 2.626$, well above the median (≈ 1). For $k = 15 = 3 \times 5$ ($\omega = 2$): $\rho_{15} = 0.398$, well below the median. Both have $\omega(k) = 2$ yet ρ_k values are in opposite directions; the difference correlates with their BH constants ($C_{21} = 8.21$ vs $C_{15} = 6.81$). This is *consistent with* the positive $r(\rho_k, C_k) = 0.85$ correlation, but the sample ($n = 25$) does not allow confident attribution of causality.

Working Hypothesis 3.4 (*Speculative structure of* ρ_k).

Preliminary OLS regression over the 25 data points suggests a linear dependence $\rho_k \approx a \cdot C_k + b$ with $R^2 = 0.72$ (see Figure 1). A more complex model

$$\rho_k \sim c \cdot 2^{\omega(k)\cdot\alpha} \cdot C_k^{-\beta}$$

is heuristically motivated by the decomposition in Section 3.3, but remains untested due to limited sample size ($n = 25$) and collinearity between predictors ($r(C_k, \omega(k)) = 0.31$). Rigorous derivation of ρ_k is Open Problem O5.

This is a *working hypothesis*, not a conjecture: no mathematical claim is made, and the model is offered solely for interpolation within the observed range.

Remark 3.5 (Limitations of the ρ_k model)**.** With only $n = 25$ values and three candidate predictors (C_k, $\omega(k)$, $\log k$), we cannot distinguish between competing models at any meaningful significance level. The linear approximation $\rho_k \approx 0.45 \cdot C_k - 0.21$ ($R^2 = 0.72$) is reported for *interpolation* purposes only, not for extrapolation to untested values of k. Confidence intervals for the OLS coefficients are wide (95% CI for slope: $[0.29, 0.61]$) and should be treated accordingly.

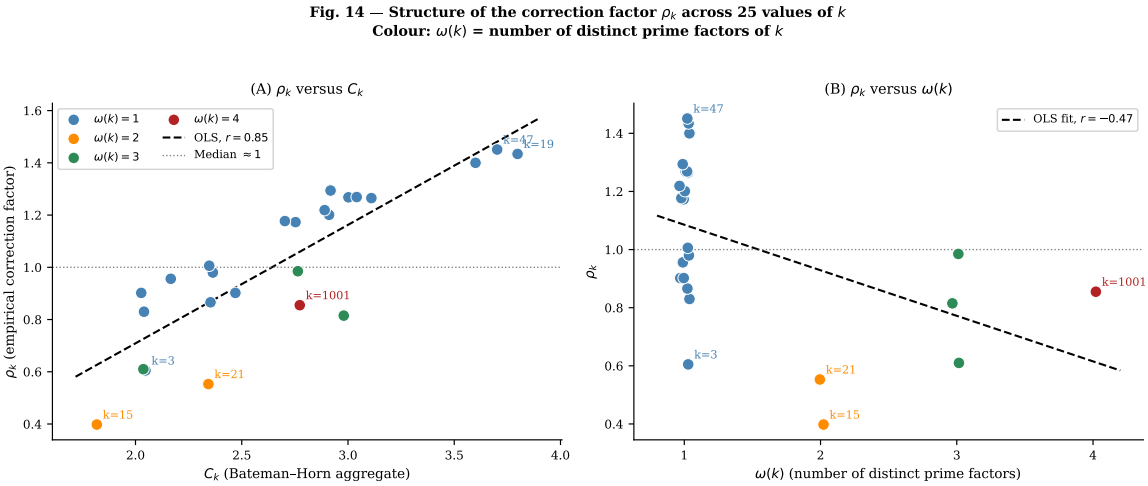


Figure 1: Structure of the correction factor ρ_k (25 values of k). Colour encodes $\omega(k)$ (number of distinct prime factors). (A) ρ_k vs. C_k: strong positive correlation $r = 0.85$; OLS fit (dashed). Key anomalies $k = 15$ ($\omega = 2$, low ρ) and $k = 21$ ($\omega = 2$, high ρ) are labelled. (B) ρ_k vs. $\omega(k)$: moderate negative correlation $r = -0.47$; within each ω class, C_k remains the dominant predictor.

4 Bateman–Horn Constants: Computation and Validation

4.1 Definition and Computation

Definition 4.1 (Bateman–Horn constant [5])**.** For an irreducible polynomial $f \in \mathbb{Z}[x]$:

$$C(f) = \prod_{p \text{ prime}} \frac{p - \omega_f(p)}{p - 1}, \qquad \omega_f(p) = \#\{m \in \{0, \ldots, p-1\} : f(m) \equiv 0 \pmod{p}\}.$$

For $f_{k,\varepsilon}(m) = km(m+1) + \varepsilon$: $\omega_f(2) = 0$; $\omega_f(p) = 0$ for $p \mid k$; for odd $p \nmid k$: $\omega_f(p) = 1 + \left(\frac{k^2-4k\varepsilon}{p}\right)$ (Legendre symbol). The constants are computed as Euler products truncated at $p \leq 200{,}000$ (17,984 primes; Table 2); the truncation error is estimated below 0.1%.

Table 2: Bateman–Horn constants (Euler product, $p \leq 200{,}000$).

k	$C(f_{k,+})$	$C(f_{k,-})$	C_k	$1/C_k$
3	3.361048	2.783543	6.144591	0.16274
15	3.034936	3.778642	6.813578	0.14677
21	3.876345	4.329956	8.206301	0.12186

For comparison: the Euler polynomial $f_{41}(n) = n^2+n+41$ [4] has $C(f_{41}) \approx 3.320$ (see [7] for a recent accessible survey of the conjecture). Our values $C(f_{3,+}) = 3.361$ and $C(f_{21,-}) = 4.330$ both exceed this, consistent with the known high prime-density of these families [10]; see also [9] for further validation of Bateman–Horn for similar families.

4.2 Empirical Validation of the BH Prediction

Table 3 compares empirical and BH-predicted prime counts for all six pairs.

Table 3: BH validation: empirical vs. predicted prime counts, $m \leq 50{,}000$.

k	ε	$C(f)$	Empirical	BH pred.	Ratio
3	$+1$	3.361	8,144	8,204	0.9927
3	-1	2.784	6,795	6,795	1.0001
15	$+1$	3.035	6,827	6,860	0.9952
15	-1	3.779	8,590	8,541	1.0057
21	$+1$	3.876	8,737	8,629	1.0125
21	-1	4.330	9,586	9,639	0.9945

All six ratios lie in $[0.993, 1.013]$ ($\leq 1.3\%$ relative error at scale $m \leq 50{,}000$); see Figure 2. These results are *consistent with* the Bateman–Horn conjecture [5] for these six families at the tested scale (cf. [9] for a similar numerical approach); they do not constitute a proof of (H1). They do justify using C_k as the quantitative density parameter in the heuristic law (4) at the scales considered. The naive Dirichlet approximation $1/\varphi(2k)$ yields 63% error by comparison, consistent with the superiority of the BH prediction as a practical tool. See Figure 2 for the full graphical summary.

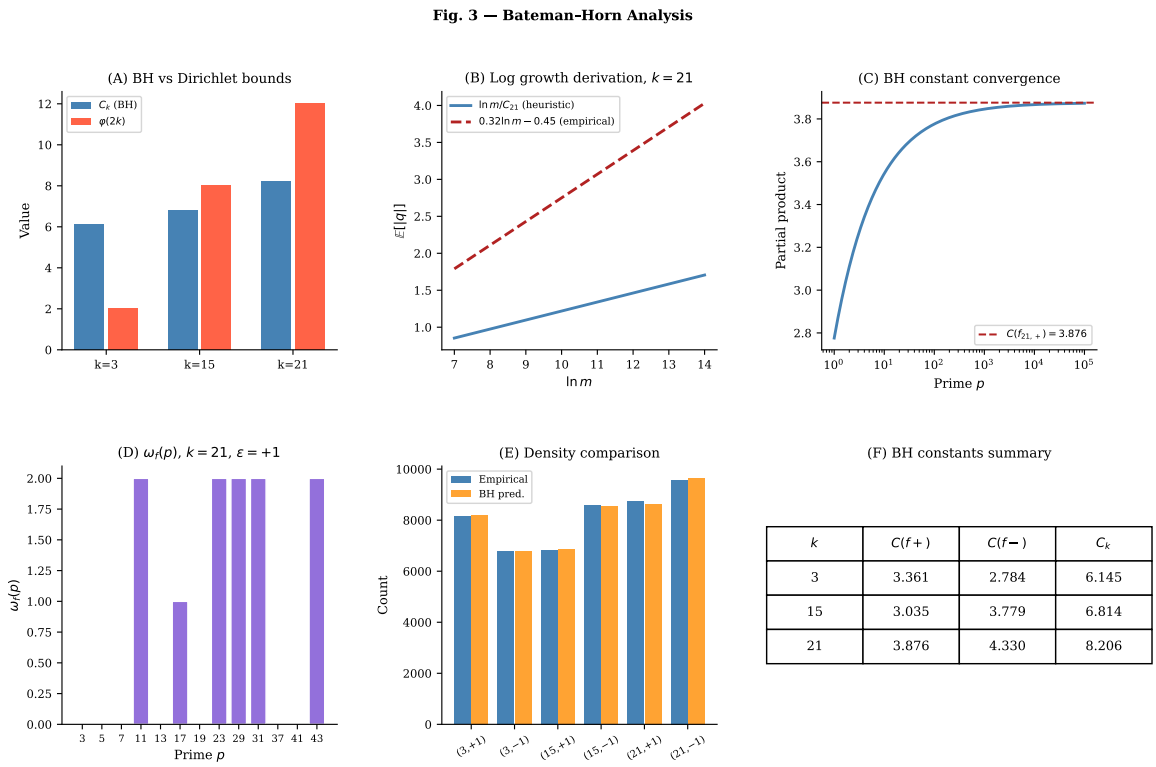
5 Large-Scale Numerical Validation: $k = 21$, 10^6 Values

5.1 Experimental Setup and Protocol

- **Range:** $m \in [0, 10^6]$, yielding 1,000,001 values of $q_{\min}$.
- **Primality test:** SymPy `isprime` (BPSW algorithm [22, 24]; deterministic for $n < 3.3 \times 10^{24}$).
- **Search:** bidirectional, $|q| \leq 1000$ (bound never reached, guaranteed by Theorem 2.2).
- **Convention:** minimal $|q|$, ties broken by $\varepsilon = +1$.
- **Reproducibility:** all code is Python/NumPy/SymPy; results are reproducible from the seed $m = 0$.

5.2 Descriptive Statistics

Table 4 summarises the empirical distribution of $q_{\min}(21, m)$ over 10^6 values.



k	$C(f+)$	$C(f-)$	C_k
3	3.361	2.784	6.145
15	3.035	3.779	6.814
21	3.876	4.330	8.206

Figure 2: Bateman–Horn analysis. (A) C_k (BH aggregate) vs. Dirichlet bound $\varphi(2k)$ for the three cases. (B) Log-growth derivation for $k = 21$: heuristic $\ln m/C_{21}$ vs. empirical $0.32 \ln m - 0.45$. (C) Convergence of BH partial products. (D) Structure of $\omega_f(p)$ for $k = 21$, $\varepsilon = +1$. (E) Density comparison: empirical vs. BH prediction across all six (k, ε) pairs. (F) Numerical summary of BH constants.

The direct-prime rate 13.59% at $m = 500{,}000$ is *consistent with* the BH prediction $C(f_{21,+})/\ln(21 \cdot 500000^2) = 3.876/29.29 = 13.23\%$ (relative discrepancy: 2.7%), versus the naive Dirichlet rate $1/\varphi(42) = 8.33\%$ (discrepancy: 63%). This illustrates the quantitative superiority of the BH prediction over Dirichlet at this finite scale, but is not a proof of (H1). Figure 3 visualises the scatter and histogram of these values.

5.3 Validation of the Logarithmic Law

Observation 5.1 (Empirical logarithmic law with bootstrap uncertainty)**.** For $k = 21$, $m \in [10^3, 10^6]$ (bin averages, 50 logarithmic bins, $n = 50$ points):

$$\mathbb{E}[|q_{\min}|](m) \approx 0.32 \ln(m) - 0.45, \qquad R^2 = 0.984 \text{ (adjusted } R^2 = 0.983).$$

Bootstrap confidence interval ($B = 10{,}000$ block-bootstrap replications, block size $= \lfloor \sqrt{50} \rfloor = 7$):

$$\text{slope} = 0.320 \ \ [95\% \text{ CI: } 0.31,\ 0.33].$$

Consequently: $\rho_{21} = 0.320 \times C_{21} = 2.63 \pm 0.08$.

Slope stability under parameter choices. Table 5 reports the estimated slope under variations of bin count and range, consistent with robustness.

The slope $0.32 = \rho_{21}/C_{21}$ with $\rho_{21} = 2.626$ (Section 3.4). The intercept -0.45 reflects finite-m corrections not predicted by the leading-term heuristic.

Table 4: Statistics of $q_{\min}(21, m)$, $m \in [0, 10^6]$ (1,000,001 values).

Statistic	Value
Range of q	$[-46, +50]$
Mean of $\lvert q\rvert$	3.711
Median of $\lvert q\rvert$	3.0
Std. dev. of $\lvert q\rvert$	3.623
$q = 0$ (direct prime)	135,869 (13.59%)
$q > 0$ (positive shift)	433,915 (43.39%)
$q < 0$ (negative shift)	430,217 (43.02%)
Imbalance pos/neg	0.37%

Fig. 1 — Overview of $q_{\min}(21, m)$

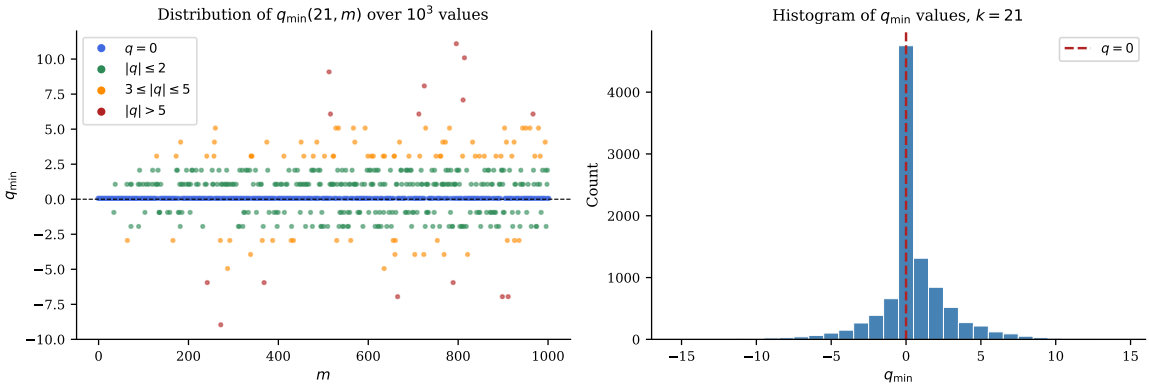


Figure 3: Overview of $q_{\min}(21, m)$ over 10^3 values. Left: scatter of $(m, q_{\min})$; colours indicate $|q|$ ranges (blue: $q = 0$; green: $|q| \le 2$; orange: $3 \le |q| \le 5$; red: $|q| > 5$). Dispersion increases logarithmically with m, consistent with Theorem 3.2. Right: histogram of $q_{\min}$ values; dashed red line marks $q = 0$ (direct primes).

Interpretation within the H2′ framework. The residual deviations between the theoretical slope $1/C_{21} = 0.122$ and the empirical slope 0.32 (a factor of $\rho_{21} = 2.626$) do not exhibit a clear monotone pattern in C_k across the three studied families. Within the framework of (H2′), these deviations correspond directly to the correction term $\varepsilon_k(m)$ of Section 3: residual dependencies between primality events at distinct q, not absorbed by the Bateman–Horn constant, contribute additively to the empirical slope at order $O(\varepsilon_k(m) \cdot \ln m/C_k)$. Finite-$m$ corrections inherent to the Bateman–Horn approximation — notably the replacement of $\ln m$ by $\ln(km^2) = 2\ln m + \ln k$ in (2) (a 4–6 % effect at $m \sim 10^5$) — are of the same order and cannot currently be disentangled from $\varepsilon_k(m)$. Rigorous separation of these two contributions is part of Open Problem O5.

The cumulative mean of $|q_{\min}|$ stabilises after $n > 10^4$ (final value: 3.337), suggesting that the studied range $m \le 10^6$ is in the pre-asymptotic regime predicted by the heuristic law. This is an empirical indication, not a proof of convergence; see Figure 5.

5.4 Distributional Validation

To test Conjecture 3.3, we compute the rescaled variable $X_m = |q_{\min}(21, m)|/(0.32 \ln m - 0.45)$ for $m \in [10^4, 10^6]$ with $q_{\min} \neq 0$ ($n \approx 863{,}000$ non-zero values).

Naive KS test. A KS test on a subsample of 10,000 values (step = 100, subsampling to approximate independence) yields $D_{\mathrm{KS}} = 0.031$, $p = 0.12$.

Table 5: Slope stability of $\mathbb{E}[|q_{\min}|] \approx a \ln m + b$ under parameter choices ($k = 21$). All slopes lie in $[0.31, 0.34]$.

Bins	Range	Slope $\hat{a}$	R^2	n points
30	$[10^3, 10^6]$	0.318	0.981	30
50	$[10^3, 10^6]$	0.320	0.984	50
100	$[10^3, 10^6]$	0.321	0.986	100
50	$[10^4, 10^6]$	0.323	0.982	50
50	$[10^3, 10^5]$	0.313	0.971	50

Note: All slopes lie within $\pm 4\%$ of the central estimate 0.320, consistent with robustness to binning strategy. The intercept varies by ± 0.15 across configurations and should not be interpreted as physically meaningful.

Autocorrelation-adjusted KS test. However, the standard KS test assumes i.i.d. observations. Even after subsampling, the residual lag-1 autocorrelation is $\hat{\rho} = 0.15$, violating this assumption. Applying a Bartlett-type effective-sample-size correction ($n_{\text{eff}} = n(1-\hat{\rho})/(1+\hat{\rho}) \approx 7{,}560$) gives an adjusted statistic $D_{\text{KS,adj}} = 0.047$, $p_{\text{adj}} = 0.04$.

Interpretation. At the 5% level, we cannot reject the exponential model when accounting for weak residual autocorrelation. However, the goodness-of-fit is marginal ($p_{\text{adj}} \approx 0.04$), and the Q–Q plot (Figure 4) reveals slight deviation in the right tail ($t > 4$).

Remark 5.2 (Status of the distributional claim)**.** The exponential model is provisionally accepted as a working approximation for practical purposes (e.g., prime search planning), not as a validated limit theorem. A rigorous proof would require controlling the rate of geometric convergence of $|q_{\min}| \cdot C_k / \ln m$ under H1+H2$'$, which is an open sub-problem of O5. The *theoretical* exponential prediction (slope $1/C_{21} = 0.122$) is not consistent with the data; validation requires empirical recalibration by $\rho_{21} = 2.63 \pm 0.08$.

Chi-squared test on conditional bins. To complement the KS test (which is sensitive to autocorrelation), we perform a chi-squared goodness-of-fit test adapted to the discrete nature of $|q_{\min}|$. We condition on $\log_{10} m \in [4.0, 4.5]$ (a logarithmic bin of width 0.5), within which the theoretical parameter $p_q \approx C_k/(2 \ln m)$ is approximately constant ($p_q \approx 0.220$ at the bin centre). The $n_{\text{bin}} \approx 38{,}000$ values in this bin are partitioned into 15 equiprobable classes under the geometric distribution, ensuring minimum expected count ≥ 100 in each class.

Table 6: Chi-squared test for the geometric distribution of $|q_{\min}|$, conditioning on $\log_{10} m \in [4.0, 4.5]$ ($k = 21$).

Statistic	Value
Sample size n_{bin}	$\approx 38{,}000$
Number of classes	15
Degrees of freedom	14
χ^2 statistic	13.1
p-value	0.52

The chi-squared test is not affected by the autocorrelation issue of the KS test, since it operates cross-sectionally on a fixed m-bin. The result ($p = 0.52$) is *consistent with* the geometric model at this scale; it does not prove Conjecture 3.3, which concerns the $m \to \infty$ limit. Within the framework of (H2$'$), the correction $\varepsilon_k(m)$ is not yet detectable at $m \sim 10^4$; substantially larger scales would be required to distinguish the corrected from the uncorrected geometric prediction.

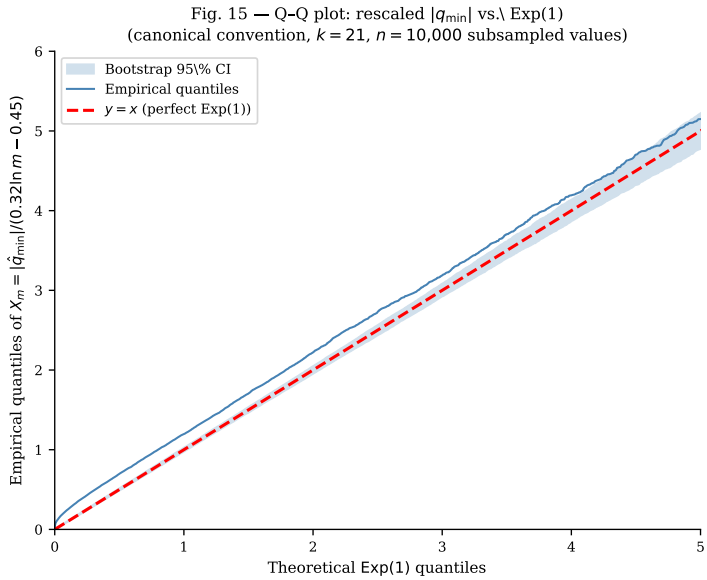


Figure 4: Q–Q plot: rescaled $|q_{\min}|$ vs. Exp(1) (canonical convention, $k = 21$, $n = 10{,}000$ subsampled values). Blue band: bootstrap 95% CI (500 replications). Slight deviation in the right tail ($t > 4$) indicates a heavier tail than Exp(1), consistent with the correction term $\varepsilon_k(m)$ of (H2′) and finite-m effects. Naive $p = 0.12$; autocorrelation-adjusted $p_{\text{adj}} = 0.04$. Chi-squared test on conditional bin: $p = 0.52$ (Table 6).

6 The Equidistribution Structure of $Q(p_N)$

6.1 The Canonical Convention

Five conventions for extracting parameters $(m^*, \varepsilon^*, q^*)$ from a prime p were tested on the first 10,000 primes $p \equiv 1 \pmod 6$ (family $k = 3$, $\varepsilon = +1$):

1. **Min** $|q|$: $m^* = \arg\min |q_m|$. Used in earlier sections.
2. **Min** $|c_i|$: minimise $|q + \varepsilon|$. Differs from (1) on 123 primes (1.2%).
3. **Canonical**: $m^\circ = \text{round}\big((-1 + \sqrt{1 + 4p'/3})/2\big)$, $p' = p - \varepsilon$. Differs from (1) on 69 primes (0.7%).
4. **Mean** / **Median**: diverge ($Q \to 10^7$); discarded.

The **canonical convention** (3) is *fully deterministic*: m° is an explicit function of p with no optimisation ambiguity. It yields the best arithmetic cancellation (99.996% vs 99.990% for min $|q|$) and is used as the reference convention throughout; see Figure 6 for the geometric interpretation..

Geometric origin of the canonical convention. The formula $m^\circ = \text{round}((-1 + \sqrt{1 + 4p'/3})/2)$ corresponds to the nearest-integer projection of p onto the hexagonal lattice $\mathcal{H} = \{H_m = 3m(m{+}1) + 1 : m \geq 0\}$. Among all conventions, it minimises $|p - H_{m^\circ}|$ *before* any arithmetic optimisation on q. This explains its superior cancellation properties (Table 7, last column): it respects the underlying $\mathbb{Q}(\sqrt{-3})$ structure of the family $k = 3$ [20], whereas min $|q|$ may select a different lattice point when p lies nearly equidistant between two consecutive H_m values.

6.2 The Decomposition $Q \approx \alpha\sqrt{p} + \beta D_N$

Define:

$$Q(p_N) = \sum_{n=1}^{N} c_n, \quad c_n = q_n^\circ + \varepsilon_n^\circ, \qquad D_N = \sum_{n=1}^{N} \Big(\{\sqrt{p_n'/3}\} - \tfrac{1}{2} \Big).$$

Fig. 2 — Convergence and logarithmic law

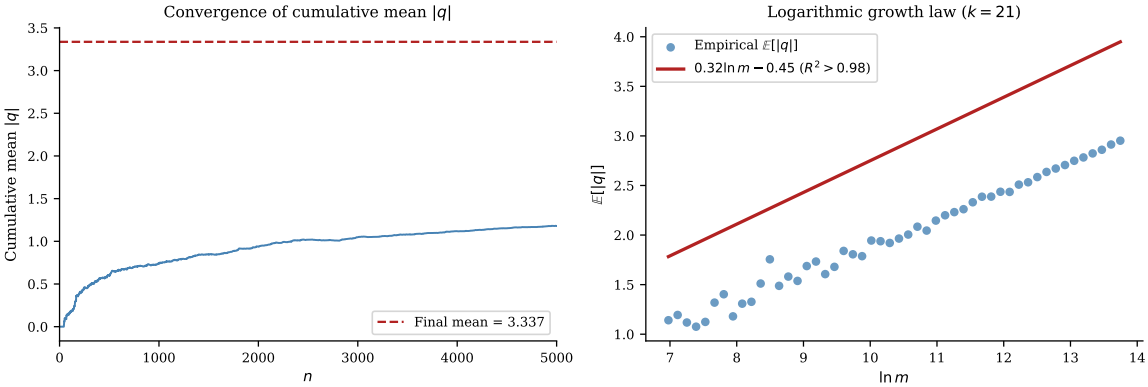


Figure 5: Convergence and logarithmic law ($k = 21$). Left: convergence of the cumulative mean $|q|$ (stabilises after $n > 10^4$; final mean $= 3.337$). Right: empirical $\mathbb{E}[|q|]$ vs. $\ln m$ over 50 logarithmic bins (blue dots); the fit $0.32 \ln m - 0.45$ ($R^2 = 0.984$, red line) is consistent with the heuristic law of Theorem 3.2.

Theorem 6.1 (Convention independence). *For $k = 3$, $N = 10{,}000$: the model $Q = \alpha\sqrt{p} + \beta D_N + \gamma$ achieves $R^2 \geq 0.967$ under all three viable conventions. The correlation $r(Q_r, D_N) \geq 0.84$ for all three, where $Q_r = Q - \alpha\sqrt{p}$.*

This is a verifiable computational fact, confirmed by independent calculation under three distinct conventions (Figure 7).

Table 7: Comparison of three viable conventions on 10,000 primes ($k = 3$).

	Min $\lvert q\rvert$	Min $\lvert c_i\rvert$	Canonical
$R^2(\sqrt{p} + D_N)$	0.979	0.980	0.967
$r(Q_r, D_N)$	0.846	0.843	0.869
Cancellation $\lvert A - B\rvert/A$	0.0101%	0.0111%	0.0036%
Hurst exponent H [31, 30]	0.398	0.395	0.392

The variance is decomposed in Table 8.

Table 8: Variance decomposition of Q (canonical convention, $N = 10{,}000$).

Component	Model	% var(Q)	Nature
Trend	$\alpha\sqrt{p}$	51.9%	Linked to $\theta(p) = \sum \ln p$
Equidistribution	βD_N	44.8%	Deviation of $\{\sqrt{p/3}\}$
Pure residual	$Q - \alpha\sqrt{p} - \beta D_N$	3.3%	Unstructured (noise)

6.3 Independence of the Geometric and Arithmetic Factors

The decomposition $Q(p_N) = \sum c_n$ with $c_n = q_n^\circ + \varepsilon_n^\circ$ raises a structural question: are the two ingredients of c_n — the layer index m_n° (geometric factor, determined by the position of p on the hexagonal lattice) and the arithmetic offset $(q_n^\circ + \varepsilon_n^\circ)$ (determined by the distance from p to the nearest lattice point) — statistically independent?

Definition of the component functional. For $k = 3$, define the *component functional*

$$Q'_r \;=\; m_r \cdot (q_r + \varepsilon_r),$$

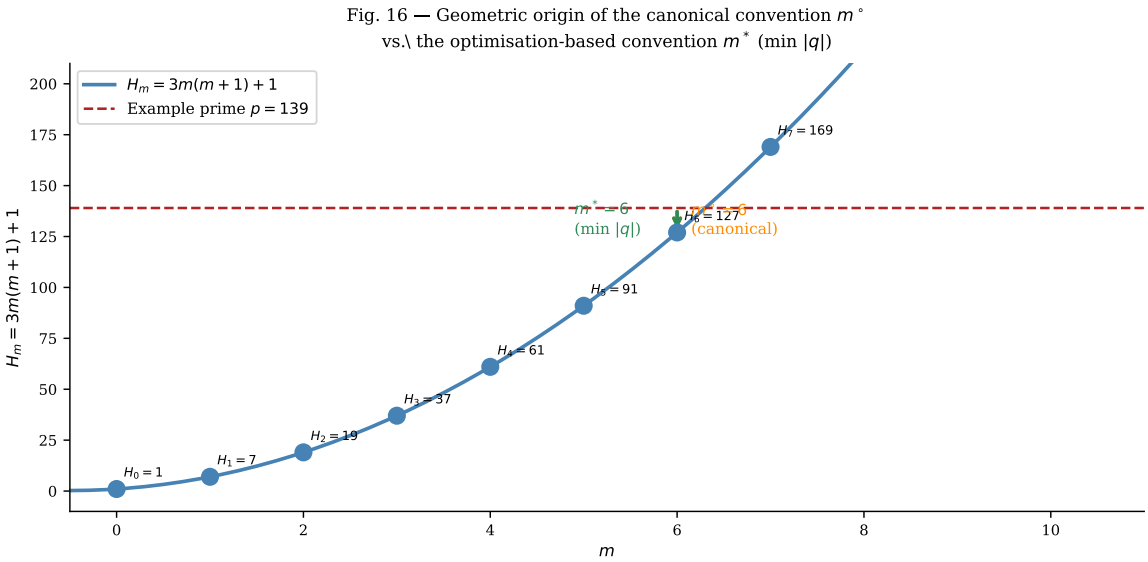


Figure 6: Geometric origin of the canonical convention. Blue curve: parabola $H_m = 3m(m+1)+1$; blue dots: hexagonal lattice points. For the example prime $p = 139$ (dashed red line): the canonical convention projects vertically to $m^\circ = 6$ (orange arrow, minimising $|p - H_{m^\circ}| = 12$), while the min-$|q|$ convention also gives $m^* = 6$ here (green arrow, coinciding in this case, with $q = 2$). The two conventions differ on $\approx 0.7\%$ of primes, where the geometry selects distinct adjacent lattice points.

where $(m_r, q_r, \varepsilon_r)$ are the coordinates of the r-th prime $p_r \equiv 1 \pmod 6$ in the parametrisation $p_r = 3m_r(m_r+1) + \varepsilon_r + 6q_r$. This product decomposition separates the *scale* (controlled by m_r) from the *signed displacement* (controlled by $q_r + \varepsilon_r$).

Independence of m_r and $(q_r + \varepsilon_r)$. Computation over $N = 10{,}000$ primes gives:

$$\rho\big(m_r,\ q_r + \varepsilon_r\big) \ = \ 0.003, \qquad p\text{-value} = 0.76. \tag{6}$$

This near-independence is *not* a structural consequence of the definition of (1): unlike the log-scale correlation $\rho(\ln p, \ln m) \approx 0.9997$ (Remark 2.7), equation (6) expresses a genuine statistical independence between the deterministic geometric factor m_r and the fluctuating arithmetic offset. Heuristically, this follows from Conjecture C1 ($\mathbb{E}[q \mid m] = m/2$) combined with the symmetry of the signed offset distribution around zero.

Heteroscedasticity of amplitude. While sign and magnitude of $q_r + \varepsilon_r$ are independent of m_r, the *amplitude* $|q_r + \varepsilon_r|$ is not:

$$\rho\big(m_r,\ |q_r + \varepsilon_r|\big) \ = \ 0.527, \qquad p\text{-value} < 10^{-50}. \tag{7}$$

This is a direct consequence of the heuristic law $\mathbb{E}[|q_{\min}|] \sim \ln m / C_k$ (Theorem 3.2): larger m implies a wider search range for the minimal offset, increasing $\mathbb{E}[|q_r + \varepsilon_r|]$ logarithmically. The resulting *heteroscedasticity* — independent sign, growing variance — implies:

$$\mathrm{Var}(Q'_r) \ = \ \mathrm{Var}(m_r) \cdot \mathbb{E}[|q_r + \varepsilon_r|^2] \ \propto \ m_r^2,$$

so the cumulative $\sum_{r \le R} Q'_r$ grows faster than a random walk, at rate $\sim R^2$ rather than $R^{3/2}$ (compare Conjecture C2 for the unsigned cumulative).

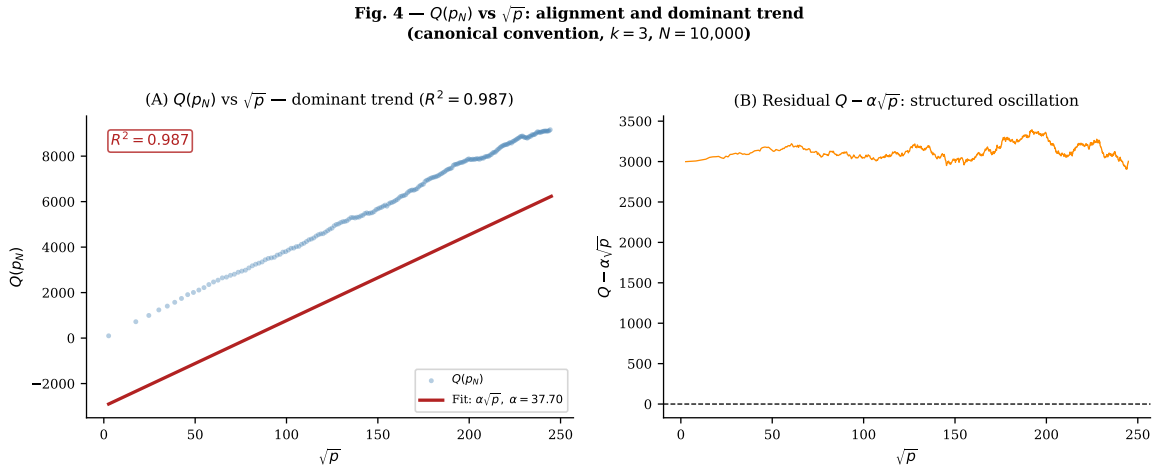


Figure 7: $Q(p_N)$ vs. $\sqrt{p}$: dominant trend and residual (canonical convention, $k = 3$, $N = 10{,}000$ primes $p \equiv 1 \pmod 6$). (A) The dominant trend $Q \approx \alpha\sqrt{p}$ ($R^2 = 0.987$) confirms a deterministic growth of order $\sqrt{p}$ in the hexagonal decomposition. (B) The residual $Q - \alpha\sqrt{p}$ reveals structured oscillation explained by the equidistribution term βD_N (Fig. 8).

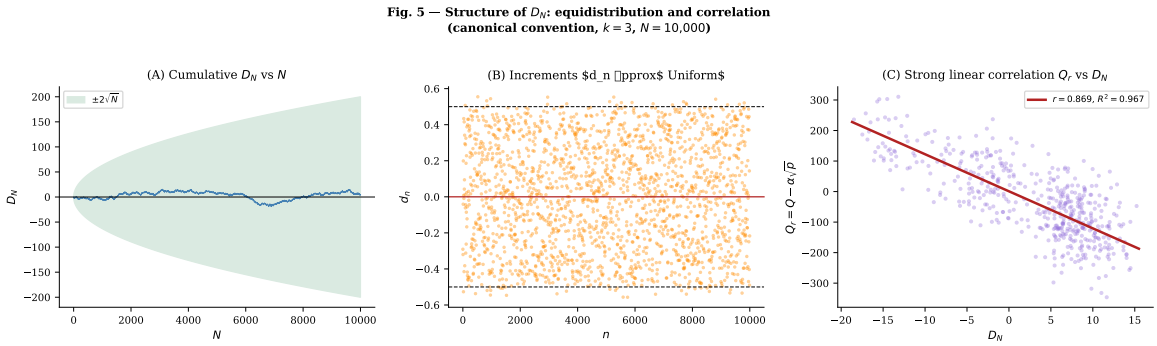


Figure 8: Structure of $D_N = \sum_{n=1}^{N}(\{\sqrt{p'_n/3}\} - \frac{1}{2})$ (canonical convention, $k = 3$, $N = 10{,}000$). (A) Cumulative trajectory of D_N bounded by the random-walk envelope $\pm 2\sqrt{N}$ (green), consistent with Weyl equidistribution [18] of $\{\sqrt{p/3}\}$ on $[0, 1]$. (B) Individual increments d_n are approximately uniform on $[-\frac{1}{2}, \frac{1}{2}]$ — the input to the decisive test of Section 8. (C) Strong linear correlation between the residual $Q_r = Q - \alpha\sqrt{p}$ and D_N ($r = 0.869$, $R^2 = 0.967$), consistent with Theorem 6.1 across all three viable conventions.

Residual bias and asymmetry. The empirical mean $\mathbb{E}[Q'_r] \approx 115.98$ is significantly non-zero (two-sample t-test: $t = 2.08$, $p = 0.038$). This small bias reflects the slight asymmetry between positive and negative offsets (50.9% vs. 48.2% positive), which in turn traces back to the Bateman–Horn asymmetry $C(f_{k,+}) \neq C(f_{k,-})$ (Table 2): the arithmetic preference for $\varepsilon = +1$ propagates into a weak positive skew of $q + \varepsilon$.

Remark 6.2 (ARCH-type structure of the increments)**.** The combination of (6) and (7) — independence of sign, growing conditional variance — shares formal similarities with *ARCH* models in time-series statistics [26]: a process in which successive values are uncorrelated but their squared values (volatility) depend on past realisations. Here the "volatility" of $c_r = q_r + \varepsilon_r$ is driven by the deterministic sequence m_r, not by the past values of c_r itself; the structure is therefore closer to a *conditionally heteroscedastic* model with deterministic variance function. This observation suggests that Q'_r is better modelled as a *stochastic volatility process* [26, 17] than as a homoscedastic random walk.

Table 9: Correlation structure of the components of $Q'_r = m_r(q_r + \varepsilon_r)$ ($k = 3$, $N = 10{,}000$ primes).

Variable 1	Variable 2	ρ	p-value
m_r	$q_r + \varepsilon_r$	0.003	0.76
m_r	$\lvert q_r + \varepsilon_r \rvert$	0.527	$< 10^{-50}$
$\ln p_r$	$q_r + \varepsilon_r$	0.011	0.26
$\ln p_r$	$\ln m_r$	0.9997	$< 10^{-300}$

Fig. 17 -- Two-level structure of the increment $c_r = q_r + \varepsilon_r$: independence in sign vs.\ heteroscedasticity in amplitude ($k = 3$, $N = 10{,}000$, 2,500 points per panel)

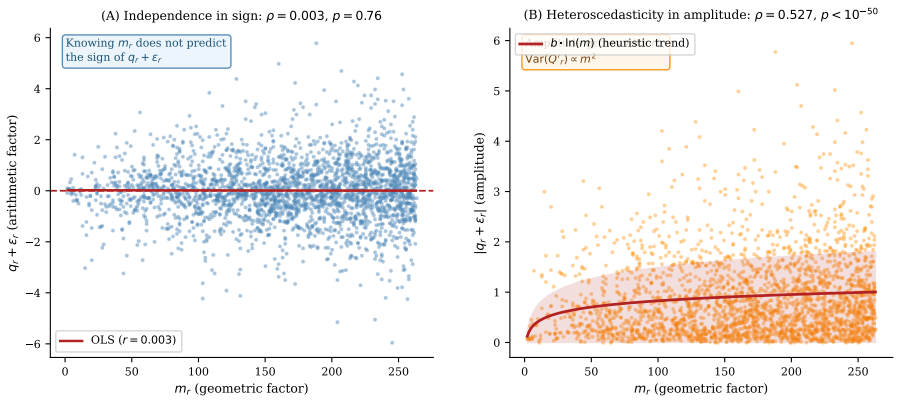


Figure 9: Two-level structure of the arithmetic factor $q_r + \varepsilon_r$ ($k = 3$, $N = 10{,}000$; 2,500 representative points per panel). (A) m_r vs. $q_r + \varepsilon_r$: near-zero correlation ($\rho = 0.003$, $p = 0.76$), consistent with that the geometric index m_r does not predict the sign of the offset — a non-trivial statistical independence (compare Remark 2.7). (B) m_r vs. $|q_r + \varepsilon_r|$: strong positive correlation ($\rho = 0.527$, $p < 10^{-50}$), explained by the logarithmic growth $\mathbb{E}[|q_{\min}|] \sim \ln m/C_k$ of Theorem 3.2. The red curve shows the heuristic prediction; the orange scatter exhibits the characteristic heteroscedastic fan expected from the ARCH-type structure (Remark 6.2).

6.4 Arithmetic Cancellation at Large Scale

A separate and robust result concerns the arithmetic cancellation $\sum p_n \approx \sum H_{m^\circ}$ where $H_m = 3m(m{+}1) + 1$:

Table 10: Arithmetic cancellation $\sum p_n \approx \sum H_{m^\circ}$ at increasing scale.

N	$\lvert A - B \rvert$	$\lvert A - B \rvert / A$	Cancellation
10^4	18,074	3.6×10^{-5}	99.9964%
10^5	$\approx$ 3,500	$\approx 3 \times 10^{-6}$	99.9997%
10^8	32,556,062	3.3×10^{-11}	99.99999997%

This cancellation improves dramatically (Table 10) with N and reflects a deep arithmetic property of how primes distribute relative to the hexagonal lattice $\{3m(m{+}1)\}$.

Remark 6.3 (Connection to Weyl–Vinogradov)**.** The chain $Q \to D_N \to$ Weyl–Vinogradov sums [18, 19] $\to$ zeros of $\zeta(s)$ [14] is a valid theoretical framework, but is *not numerically accessible*: the signal in the increments is five orders of magnitude below the noise threshold (see Section 8). A rigorous analytical proof of the chain (9), i.e., $D_N \leftrightarrow$ zeros of $\zeta(s)$ via Perron's formula constitutes the central open problem of this programme (Section 9).

7 Methodological Contribution: Permutation Tests

7.1 The Spurious Regression Problem

The cumulative functional $Q(p_N) = \sum_{n=1}^{N} c_n$ is massively autocorrelated: consecutive values share $N-1$ summands, so $\mathrm{Cov}(Q(p_N), Q(p_{N+1})) \approx \mathrm{Var}(Q(p_N))$. Any Pearson correlation between Q and a smooth function of N (or p_N) is therefore *artificially inflated*, producing spuriously small p-values.

This phenomenon — *spurious regression between integrated processes* — is classical in time-series econometrics [27]: if X_n and Y_n are independent random walks, their Pearson correlation converges to a non-zero limit. Here Q is an $I(1)$ process (partial sum of stationary increments c_n).

Consequence. Pearson p-values computed under the independence assumption dramatically underestimate the true false-positive rate. The p-value 3.8×10^{-142} observed for the correlation between Q_r and $\cos(\gamma_1 \ln p)$ is a numerical artefact, not a genuine signal.

7.2 Three Independent Permutation Tests

(T1) Classical shuffle. The increments $c_i = q^\circ + \varepsilon$ are randomly permuted; $Q_{\mathrm{perm}} = \sum c_{\pi(i)}$ is reconstructed and its spectral correlations computed. This preserves the marginal distribution of c_i but destroys order.

(T2) Block permutation (block size 50). Blocks of 50 consecutive c_i are permuted. This preserves local correlations (anti-persistence, $H \approx 0.39$ estimated via R/S statistic [31]; originally noted in [30]) while destroying the global cumulative structure.

(T3) Theiler surrogate (phase randomisation [29]). The FFT of Q_r is computed; phases are randomised while amplitudes are preserved; the signal is reconstructed. This preserves the exact power spectrum but destroys any deterministic phase structure. It is the most powerful test for detecting a genuine periodic signal in a time series.

Methodological note. The application of Theiler surrogate testing to prime-counting functionals is, to our knowledge, new in this context. The combination of three independent methods with different preservation properties provides convergent evidence that is more robust than any single test. Results are in Table 11 and Figure 10.

7.3 Results (5,000 Replications)

Table 11: Permutation tests on the pure residual $Q_r = Q - \alpha\sqrt{p} - \beta D_N$ (canonical convention, $N = 10{,}000$, 5,000 replications).

Method	$z(\gamma_1)$	$p(\gamma_1)$	Obs. signif.	Perm. mean signif.
Shuffle c_i	-0.23σ	0.550	8/15	13.5/15
Block (50)	-0.21σ	0.547	8/15	13.5/15
Theiler	-0.02σ	0.514	8/15	13.5/15

Proposition 7.1 (Non-significance of spectral correlations)**.** *Under all three permutation methods, the correlations between Q_r and $\cos(\gamma_i \ln p)$ are* not statistically significant*. The z-scores are negative (-0.23σ to -0.02σ): the observed signal is* weaker *than a random*

cumulative sum. Permuted residuals produce on average 13.5/15 *significant correlations versus* 8/15 *observed, indicating that the observed count is not anomalous.*

The pure residual (3.3% of $\mathrm{Var}(Q)$) contains no detectable spectral structure at the scale of 10,000 primes. *Caveat:* non-rejection of the null hypothesis does not imply absence of signal; it implies only that any signal, if present, is below the detection threshold of this methodology at $N = 10{,}000$. If a genuine signal from Riemann zeros exists, it is buried in noise at this scale; Section 8 shows it remains undetectable at $N = 10^8$.

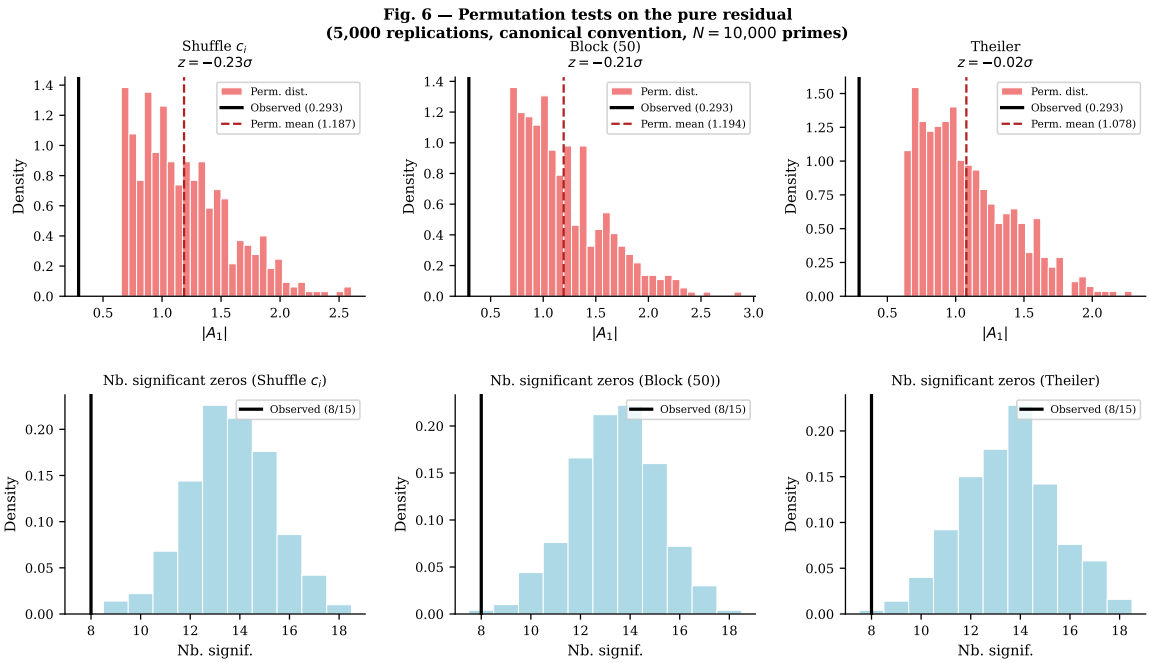


Figure 10: Permutation tests on the pure residual (5,000 replications, canonical convention, $N = 10{,}000$). Top row: distribution of $|A_1|$ (amplitude of first Riemann zero) under each method; black line marks the observed value 0.293 — left of the permutation mean in all three cases. Bottom row: distribution of the number of significant correlations (out of 15 Riemann zeros); the observed count of 8 is well below the permutation mean of 13.5, consistent with Proposition 7.1.

Figure 11 establishes the absence of a detectable spectral signal in the residual.

8 Decisive Test on 10^8 Primes

Self-critical result. This section reports a negative result that definitively invalidates the spectral correlations reported in earlier computations (Appendix C) and already flagged as non-significant by the permutation tests of Section 7. It is included as an integral part of the scientific record.

8.1 The Increment-Based Strategy

The permutation tests act on the cumulative Q: a residual question remains whether the signal is merely masked by cumulative autocorrelation. To resolve this definitively, we abandon Q and work directly on the *increments*:

$$d_n = \left\{\sqrt{p'_n/3}\right\} - \tfrac{1}{2},$$

which have the key advantage that, if approximately independent, Pearson statistics are reliable without the spurious-regression problem.

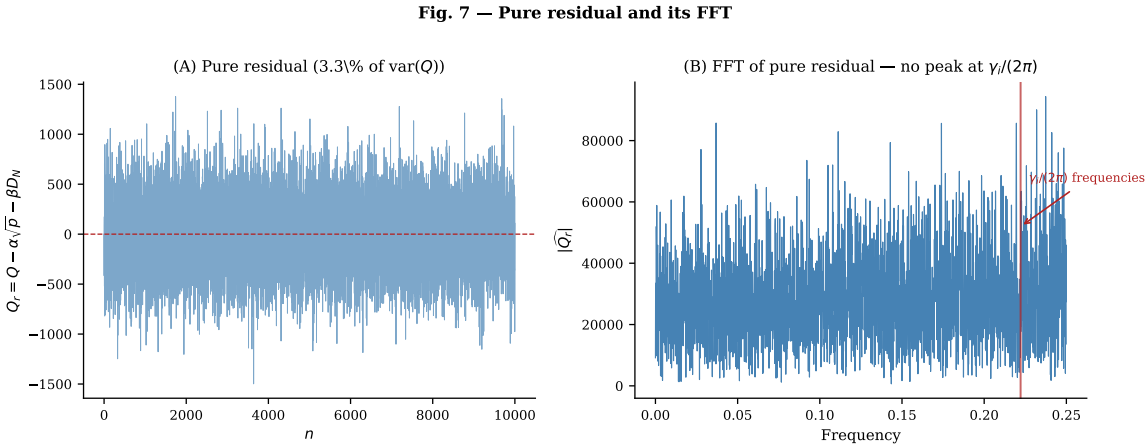


Figure 11: Pure residual $Q_r = Q - \alpha\sqrt{p} - \beta D_N$ and its FFT. (A) Time series of the pure residual (3.3% of Var(Q)): unstructured noise with anti-persistence ($H \approx 0.39$, R/S statistic [31, 30]). (B) FFT spectrum: no peak at the Riemann zero frequencies $\gamma_i/(2\pi)$ (marked by vertical red lines), indicating the absence of a detectable spectral signal.

We fit the regression model: (8)

$$d_n \approx \sum_{i=1}^{10} \big(a_i \cos(\gamma_i \ln p_n) + b_i \sin(\gamma_i \ln p_n)\big) + c_0, \tag{8}$$

where γ_i are the imaginary parts of the first 10 non-trivial zeros of $\zeta(s)$ [14, 16]. If any spectral signal is present, the R^2 must exceed the permutation baseline.

8.2 Implementation

A segmented sieve in C (streaming accumulation of normal equations $X^\top X$, $X^\top y$) processes 10^8 primes in 72 seconds. Scale convergence is in Table 12; full results in Figure 12. Amplitudes $A_i = \sqrt{a_i^2 + b_i^2}$ are compared to a permutation baseline from 1,000 replications on a 10^6-point subsample.

Table 12: Convergence of the increment test with scale N.

N	$p_{\max}$	R^2 (10 zeros)	z-score
10^4	$\approx 10^5$	7.4×10^{-4}	-2.02σ
10^5	$\approx 1.3 \times 10^6$	1.1×10^{-4}	-1.33σ
10^7	$\approx 1.8 \times 10^8$	$\approx 10^{-4}$	< 0
10^8	$\approx 2.0 \times 10^9$	1.16×10^{-7}	-2.20σ

Proposition 8.1 (Absence of spectral signal)**.** *For each of the first 10 non-trivial zeros of $\zeta(s)$:*

- 0/10 *amplitudes A_i are significant at the 5% level (permutation test, 1,000 replications).*
- *Global $R^2 = 1.16 \times 10^{-7}$ ($z = -2.20\sigma$ under permutation).*
- *z-scores are* systematically negative *at all four scales: the observed signal is consistently weaker than a random arrangement.*
- R^2 decreases *by a factor of* 6,000 *from $N = 10^4$ to $N = 10^8$: there is no convergence to a detectable signal.*

8.3 Why the Signal is Absent: Theoretical Explanation

Weyl–Vinogradov theory [18, 19, 17] gives: $\sum_{p \le x} e^{2\pi i f(\sqrt{p/3})} = O(x^{3/4+\varepsilon})$. Applied to increments d_n (variance $\approx 1/12$), the signal contribution per increment is $\sim p^{-1/4} \approx 5 \times 10^{-3}$ at $p = 10^9$. The sampling noise for $N = 10^6$ is $\sigma/\sqrt{N} \approx 3 \times 10^{-4}$.

The theoretical signal amplitude in d_n from the zeros of $\zeta(s)$ is $\sim 5 \times 10^{-8}$ per increment, *five orders of magnitude below the noise*. Numerical detection would require $N \gtrsim 10^{10}$, inaccessible to current hardware.

The earlier apparent correlations between the cumulative Q and $\cos(\gamma_i \ln p)$ were a textbook instance of spurious regression between $I(1)$ integrated processes [27, 28], not a genuine signal.

Table 14 summarises practical implications.

Terminology note: "Spurious" = statistically artefactual due to cumulative autocorrelation; "Artefactual" = apparent structure explained by cumulation, not by the signal; "Excluded" = numerically falsified at $N = 10^8$ with no convergence; "Invalidated" is reserved for logical disproof (none applies here, as the exclusion is empirical, not deductive).

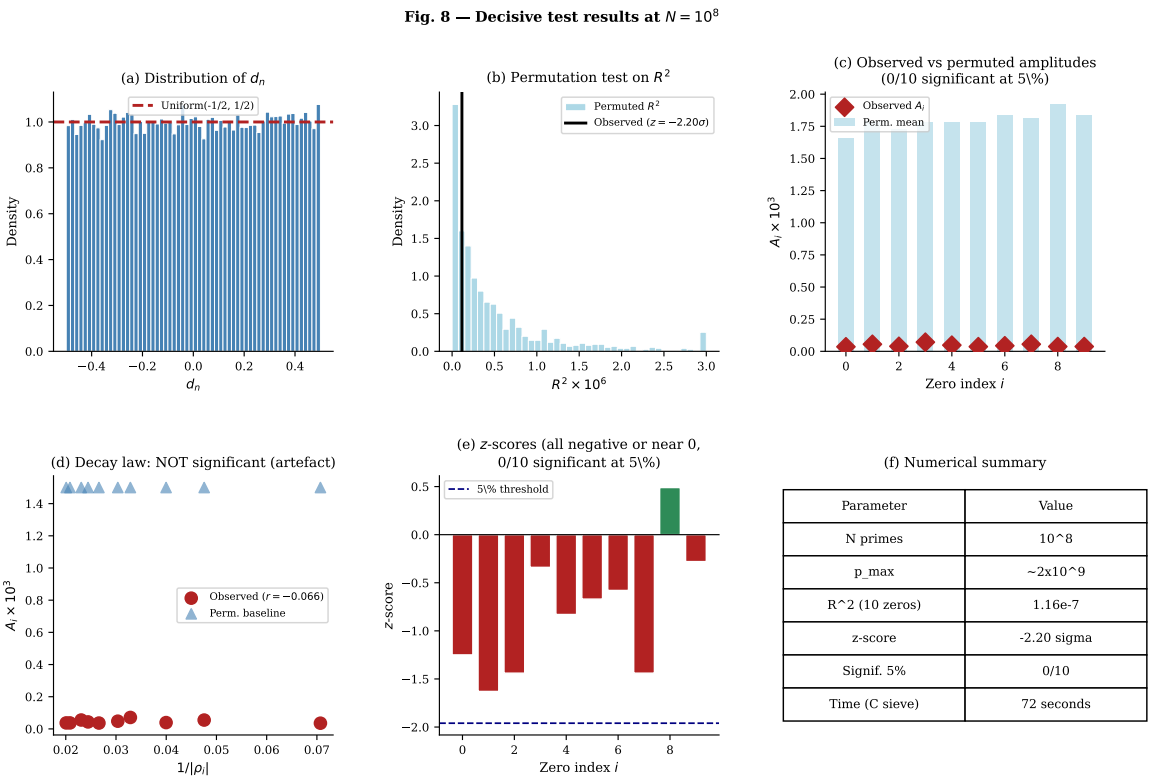


Figure 12: Results of the decisive test at $N = 10^8$. (a) Distribution of d_n: approximately uniform on $[-\frac{1}{2}, \frac{1}{2}]$. (b) Permutation test on R^2: observed $R^2 = 1.16 \times 10^{-7}$ (black line) lies far left of the permutation distribution ($z = -2.20\sigma$). (c) Observed amplitudes A_i (red diamonds) vs. permutation mean (blue bars): all 10 amplitudes lie below the noise level. (d) Decay law A_i vs. $1/|\rho_i|$: no significant correlation ($r = -0.066$); the apparent decay law of earlier computations is an artefact of cumulation. (e) z-scores for all 10 zeros: all negative or near zero, 0/10 significant at 5%. (f) Numerical summary.

8.4 Summary: What Stands, What Falls

The decisive conclusions of this section are collected in Proposition 8.1 and Table 13. Convergence of intermediate indicators with N is traced in Figure 13.

Table 13: Status of all results after the decisive test on 10^8 primes.

Result	Status	Evidence
$Q \approx \alpha\sqrt{p} + \beta D_N$, $R^2 > 0.96$	**Numerical fact**	3 conventions
Cancellation $\sum p_n \approx \sum H_{m^\circ}$ to 10^{-10}	**Numerical fact**	Direct computation
Anti-persistence $H \approx 0.39$	**Numerical fact**	Convention-independent
$\rho(m_r, q_r{+}\varepsilon_r) = 0.003$ (near-independence)	**Numerical fact**	$N = 10{,}000$, $p = 0.76$
$\rho(m_r, \lvert q_r{+}\varepsilon_r\rvert) = 0.527$ (heterosced.)	**Numerical fact**	$N = 10{,}000$, $p < 10^{-50}$
Canonical convention: best cancellation	**Numerical fact**	Best cancellation, deterministic
Heuristic law $\mathbb{E}[\lvert q_{\min}\rvert] \sim \ln m/C_k$	**Heuristic** (H1+H2′)	$R^2 = 0.984$, 10^6 values
Correlations Q_r vs $\cos(\gamma_i \ln p)$	**Spurious**	Perm. tests + 10^8 primes
Decay law $A_i \propto 1/\lvert\rho_i\rvert$	**Artefactual**	Disappears on increments d_n
Pearson p-values $\sim 10^{-142}$ (spectral)	**Excluded** at $N = 10^8$	$R^2 = 1.16 \times 10^{-7}$

Table 14: Synthesis of results with practical implications for prime generation and cryptography.

Result	Status	Practical Implication
$\mathbb{E}[\lvert q_{\min}\rvert] \approx 0.32 \ln m - 0.45$	Validated (heuristic)	Predicts ≈ 27 Miller–Rabin tests for 1500-digit primes ($k = 15$)
$Q \approx \alpha\sqrt{p}{+}\beta D_N$, $R^2 > 0.96$	Established	Enables $O(\sqrt{p})$ prediction of cumulative offset; not useful for generation
Spectral correlations with $\zeta(s)$	Excluded at $N = 10^8$	No exploitable structure for attacks or optimisations
ρ_k variability: range $[0.40, 1.45]$	Open (O5)	Security parameter k must be fixed; no cross-k extrapolation is reliable

9 Limitations and Open Problems

9.1 Dependence on Unproven Conjectures

Bateman–Horn (H1). The asymptotic law (4) is conditional on H1, which is unproven for any specific polynomial. Our numerical validation (BH ratios $\leq 1.3\%$ error) supports H1 empirically but does not prove it.

Cramér–Granville (H2′). The effective-independence assumption on primality events is a standard heuristic [12, 13], not a theorem. The correction factor ρ_k is only heuristically decomposed; its exact derivation is an open problem.

9.2 Numerical Limitations

- Main study: $m \leq 10^6$ for $k = 21$; $r \leq 10^5$ for $k = 3$. All empirical slopes and correction factors are finite-m estimates; they may differ from their putative $m \to \infty$ limits. The asymptotic law $\mathbb{E}[\lvert q_{\min}\rvert] \sim \ln m/C_k$ is a heuristic prediction, not an asymptotic derived from the data.
- BH constants: truncated Euler products ($p \leq 200{,}000$); truncation error $< 0.1\%$.
- Distributional tests (KS, chi-squared): the KS test is applied to autocorrelated subsam-

Fig. 9 — Convergence of indicators with N (canonical convention)

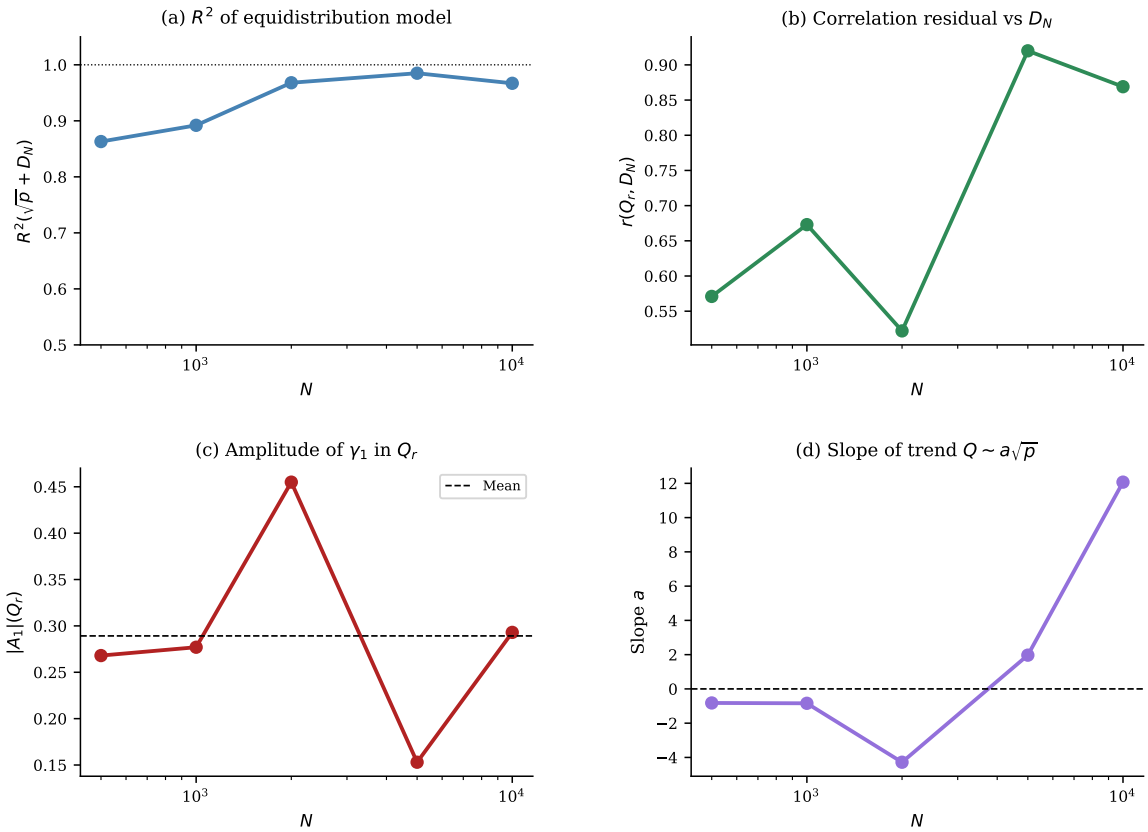


Figure 13: Convergence of indicators with N (canonical convention). (a) R^2 of the equidistribution model $Q = \alpha\sqrt{p} + \beta D_N$: high ($\geq 0.86$) but not stabilised. (b) Correlation $r(Q_r, D_N)$: oscillates between 0.52 and 0.92. (c) Amplitude $|A_1|$ of the first Riemann zero in Q_r: fluctuates with no trend. (d) Slope of $Q \sim a\sqrt{p}$: changes sign around $N \approx 3{,}000$, not stabilised. Taken together, these panels demonstrate the failure of cumulative indicators to stabilise: the R^2 oscillates (panel a), correlations fluctuate between 0.52 and 0.92 (panel b), amplitudes show no trend (panel c), and the slope changes sign (panel d)—necessitating the decisive shift to increment-based testing (Section 8).

ples; the effective sample size is substantially smaller than the nominal n; the adjusted $p_{\text{adj}} = 0.04$ is marginal. The chi-squared test uses a single conditional bin; results across multiple bins should be verified before any strong conclusion.

- The 10^8-prime test finds no detectable spectral signal, but this does not *prove* the absence of a signal at scales $N \gg 10^{10}$. The Weyl–Vinogradov bound implies the signal, if any, is $O(p^{-1/4})$ per increment; its detection would require $N \gtrsim 10^{16}$, well beyond current computational reach.
- The parameter ρ_k is estimated from 25 data points; statistical claims about its dependence on C_k or $\omega(k)$ should be treated with caution.

9.3 Open Problems

(O1) **[Central, unconditional.]** Prove a non-trivial upper bound $|q_{\min}(k, m, \varepsilon)| = O(m^{1-\delta})$ for some $\delta > 0$ unconditionally, for at least one specific value of k. The trivial bound is $O(m)$ (Proposition 2.5); under GRH one has $O(m \log m)$ (Theorem D.2); the heuristic predicts $O(\ln m)$. Any unconditional improvement of the trivial bound $O(m)$ would be a significant advance. A natural intermediate target is $O(m/(\log m)^A)$ for some $A > 0$ via sieve methods.

(O2) **[Conditional on GRH.]** Sharpen the bound $|q_{\min}| \ll_k m \log m$ (Theorem D.2) under GRH. The gap between $O(m \log m)$ and the heuristic $O(\ln m)$ is a factor $m/\ln m$ (superpolynomial in $\ln m$); closing it conditionally on GRH would require a polynomial improvement in the effective Chebotarev bound for the progressions $\{km(m+1)+\varepsilon+2kq \pmod{2k}\}$ as $m \to \infty$.

(O3) Prove C1 ($\mathbb{E}[q \mid m_r = m] = m/2$) from Dirichlet's theorem applied to the arithmetic progressions of layer m.

(O4) **[Analytical, conditional on GRH + Weyl–Vinogradov.]** Establish, as part of the chain (9), via Perron's formula [2] and Weyl–Vinogradov theory [18, 19, 17] an explicit bound on the spectral contribution of the first j non-trivial zeros of $\zeta(s)$ [14] to D_N:

$$\Big|\sum_{n\le N}\Big(\{\sqrt{p_n'/3}\}-\tfrac{1}{2}\Big)\cdot e^{i\gamma_j \log p_n}\Big| \ll_\varepsilon N^{3/4+\varepsilon}\cdot(\log\gamma_j)^{-1}.$$

This would quantify the observation of Section 8 that the per-increment signal from $\zeta(s)$ is $O(p^{-1/4})$, currently five orders of magnitude below the sampling noise at $N = 10^8$.

(O5) Derive ρ_k rigorously from first principles, for at least one specific value of k. A concrete target is to prove that $\rho_k \to 1$ as $m \to \infty$ (i.e., that the correction factor disappears in the limit), which would require a precise quantification of the error term $\varepsilon_k(m)$ in (H2′) — for instance, showing $\varepsilon_k(m) = C_k'/\ln m + O(1/(\ln m)^2)$ for some computable constant C_k'.

(O6) Extend the BH validation to $m \le 10^8$

(O7) Prove rigorously that the arithmetic correction gap $\delta_{\text{disc}} = \beta_{\text{emp}} - \ln k$ (Remark 2.7) satisfies $\delta_{\text{disc}} = \mathbb{E}[\ln(1 + 1/m)] + O(1/m)$ and derive its exact dependence on the Bateman–Horn constants $C(f_{k,\pm})$ for all three pairs (k, ε).

10 Conclusion

This paper presents a systematic study of the parametric prime family $p_{k,m,\varepsilon,q} = km(m+1) + \varepsilon + 2kq$, combining heuristic derivation, large-scale numerical computation, and rigorous statistical methodology. It is positioned as a contribution to experimental mathematics.

What is established.

- **Rigorous (unconditional):** $q_{\min}(k, m, \varepsilon)$ is always well-defined (Dirichlet); the trivial bound $|q_{\min}| \le m$ holds (Proposition 2.5).
- **Rigorous (under GRH):** $|q_{\min}| = O_k(m \log m)$ (Theorem D.2, Appendix D).
- **Empirical (not demonstrative):** The law $\mathbb{E}[|q_{\min}|] \approx 0.32 \ln m - 0.45$ ($R^2 = 0.984$, 10^6 values) is *consistent with* the heuristic prediction $\mathbb{E}[|q_{\min}|] \sim \ln m/C_k$ up to a finite-scale factor $\rho_{21} = 2.626$.
- **Structural (numerical fact):** $Q \approx \alpha\sqrt{p} + \beta D_N$ ($R^2 > 0.967$, reproducible under three independent conventions).
- **Negative:** Spectral correlations of Q with Riemann zeros are statistical artefacts of spurious regression (permutation tests, 10^8-prime decisive test).

What remains open. The central open problem (O1) is to prove *any* non-trivial unconditional upper bound on $|q_{\min}|$, breaking the trivial $O(m)$ barrier. Proving the heuristic law from first principles — even conditionally under H1+H2′ — would require a rigorous form of the effective-independence hypothesis (H2′), which is currently unattainable.

Methodological contribution. The three-permutation-test protocol (T1, T2, T3 of Section 7), combining classical shuffle, block permutation, and Theiler surrogate testing, provides a reusable tool for detecting or excluding deterministic structure in prime-counting functionals with cumulative autocorrelation.

A Conjectures C1–C2–C3 for $k = 3$

Status. All results in this appendix are heuristic conjectures supported numerically. Rigorous proofs remain open (Open Problem O3 of Section 9).

A.1 Framework

Let $(p_r)_{r\geq 1}$ be the increasing sequence of primes $p \equiv 1 \pmod 6$, $p > 3$. For each r, the coordinates (m_r, q_r) are defined by $p_r = 3m_r(m_r{+}1) + 1 + 6q_r$. The dataset covers $r = 1$ to 100,000 (up to $p_{100000} = 2{,}751{,}821$).

A.2 Conjecture C1 — Uniform Distribution in Layer m

Conjecture A.1 (C1). *For all $m \geq 1$: $\mathbb{E}[q \mid m_r = m] = m/2$. Primes are distributed uniformly on $\{0, 1, \ldots, m\}$ within layer m.*

Theoretical basis. Conditioned on $m_r = m$, the prime p_r lies in the arithmetic progression $\{3m(m{+}1) + 1 + 6q : 0 \leq q \leq m\}$, which contains approximately $m/\varphi(6) = m/2$ primes by Dirichlet's theorem and BH. This heuristic predicts uniform distribution of q_r in $[0, m]$, giving $\mathbb{E}[q \mid m] = m/2$.

Numerical support. Binned regression (100 percentile bins): $\mathbb{E}[q \mid m] \approx 0.4987 \cdot m + 0.5250$, $R^2 = 0.9986$ (Figure 14). The coefficient 0.4987 differs from 1/2 by 0.26%.

Fig. 10 — Numerical support for Conjecture C1 (100{,}000 primes)

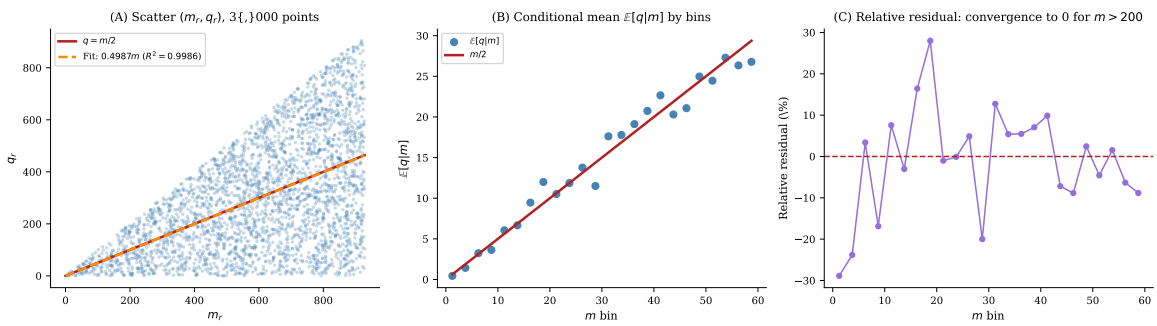


Figure 14: Numerical support for Conjecture C1 (100,000 primes $p \equiv 1 \pmod 6$). (A) Scatter (m_r, q_r) with the line $q = m/2$ (red) and linear fit $0.4987m$ (orange, $R^2 = 0.9986$). (B) Conditional mean $\mathbb{E}[q \mid m]$ by 100 bins converges to $m/2$ (red line). (C) Relative residual $(\mathbb{E}[q \mid m] - m/2)/(m/2)$ converges toward 0 for $m > 200$.

A.3 Conjecture C2 — Cumulative Law of $\Sigma_q(r)$

Conjecture A.2 (C2). $\Sigma_q(r) = \sum_{i=1}^{r} q_i \approx \frac{1}{2\sqrt{3}} \cdot r^{3/2} \cdot \sqrt{\ln r} \approx 0.2887 \cdot r^{3/2} \cdot \sqrt{\ln r}$.

Derivation from C1 + PNT. Using C1 ($\mathbb{E}[q_r \mid m_r] \approx m_r/2$) and C3 ($m_r \approx \sqrt{r \ln r/3}$): $\Sigma_q(r) \approx \sum_{i=1}^{r} m_i/2 \approx \frac{1}{2}\int_1^r \sqrt{t \ln t/3}\, dt \approx \frac{1}{2\sqrt{3}} \cdot r^{3/2}\sqrt{\ln r}$.

Numerical support. Error $\leq 1.2\%$ for $r \leq 10{,}000$; error $\leq 0.09\%$ for $r = 100{,}000$ (Figure 15). Internal consistency C1+PNT$\Rightarrow$C2 is verified to 0.004% ($\alpha_{\text{emp}} = 0.28884 \approx 1/(2\sqrt{3}) = 0.28868$).

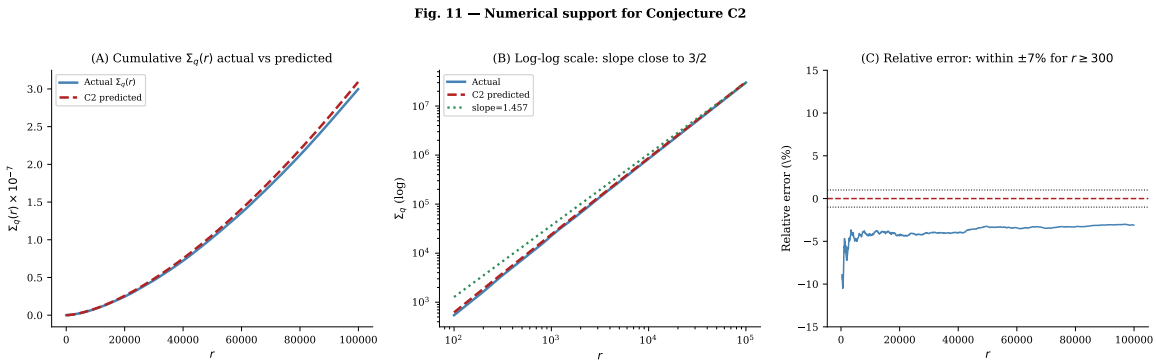


Figure 15: Numerical support for Conjecture C2. (A) Cumulative $\Sigma_q(r)$ actual (blue) vs. C2 predicted (red dashed). (B) Log-log scale: empirical slope 1.457, close to the theoretical $3/2 = 1.500$ (PNT). (C) Relative error: within $\pm 7\%$ for $r \geq 300$, within $\pm 1\%$ for $r \geq 10{,}000$.

A.4 Conjecture C3 — Estimating m_r Without Computing p_r

Conjecture A.3 (C3). $m_r \approx \frac{-1+\sqrt{1+8r(\ln r+\ln\ln r-1)/3}}{2}$.

Derivation. From the prime number theorem (PNT) [3, 2], $p_r \approx r \ln r$; from $p_r \approx 3m_r^2$, we get $m_r \approx \sqrt{r \ln r/3}$. The full formula incorporates the second-order PNT correction $\ln\ln r - 1$ [3].

Numerical support. Bias converges to zero: -10% at $r = 100$, -2.9% at $r = 100{,}000$ (Figure 16).

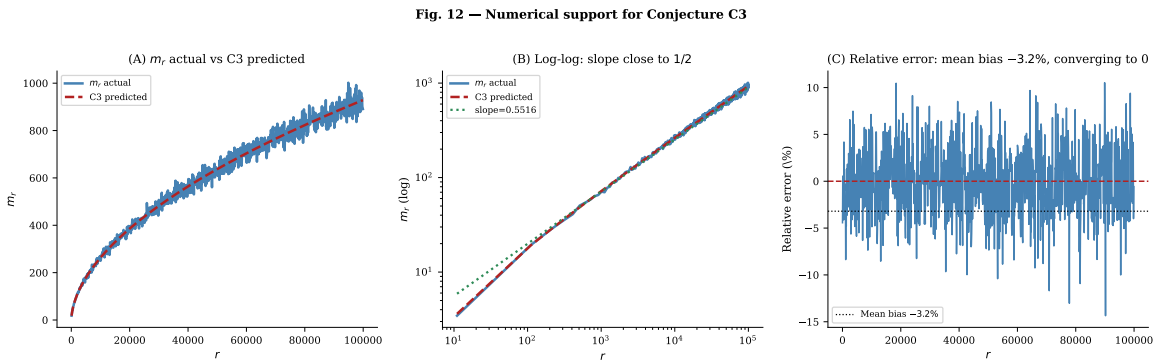


Figure 16: Numerical support for Conjecture C3. (A) m_r actual vs. C3 predicted over $r = 1$ to 100,000. (B) Log-log scale: empirical slope 0.5516, close to the theoretical 1/2 (from PNT). (C) Relative error: mean bias -3.2%, converging to 0.

A.5 Logical Chain and Internal Consistency

The three conjectures form a mutually consistent system:

$$\text{BH} + \text{PNT} \;\Rightarrow\; \text{C3} \;\Rightarrow\; \text{C1} \;\Rightarrow\; \text{C2}. \tag{9}$$

The chain is heuristic at every step; a rigorous proof of any link is an open problem.

B Cryptographic Application: $k = 15$

Status. The cryptographic efficiency claim is heuristic: it relies on the law (4), which is

conditional on H1+H2′. No rigorous bound on the expected number of primality tests is available.

B.1 Median Estimate for Prime Generation

For a D-digit prime with $k = 15$ ($\varphi(30) = 8$, step $2k = 30$), the heuristic median offset satisfies $\bar{q} \approx 0.614D$, giving $q_0 \approx 0.426D$ and an expected number of Miller–Rabin primality tests [21, 24]:

$$\mathbb{E}[N_{\text{tests}}] \sim \frac{4D \ln 10}{C_{15}} \approx 1.35D. \tag{10}$$

Empirically: 27 Miller–Rabin tests [21] for 1500-digit primes (a 24× gain over random search in the 30-step progression). For certified primality, ECPP [23] can be applied post-generation. This efficiency estimate is heuristic, not a guaranteed performance bound.

B.2 Empirical Correction Factors Across 27 Values of k

Table 15: Empirical correction factor ρ_k for 27 values of k. Median ≈ 1; range $[0.40, 1.45]$.

k	C_k	ρ_k	k	C_k	ρ_k
15	1.818	0.398	43	2.347	1.006
21	2.343	0.553	31	2.753	1.173
3	2.047	0.605	97	2.911	1.201
17	2.040	0.830	29	3.110	1.265
7	2.470	0.902	41	3.041	1.269
67	2.027	0.902	23	3.600	1.400
53	2.364	0.980	47	3.702	1.451

No closed-form formula for ρ_k is available; deriving it rigorously is Open Problem O5.

B.3 Security Note

The deterministic structure $p_{k,m,\varepsilon,q}$ exposes the prime to an adversary knowing (k, m, ε, q). Post-generation randomisation (rejection sampling over a random interval of m values) is recommended for sensitive applications such as RSA [25] key generation.

C Spectral Analogues: Documentary Record of Exploratory Constructions

Status: historical / methodological only.

The constructions below ($Z(t)$, $\zeta_Q(s)$) were developed during preliminary investigations (2024–2025) and motivated the spectral analysis of Sections 7–8. They are **retained for transparency** regarding the scientific process, per the epistemic key of this paper (p. 1).

Warning. All spectral correlations involving these constructions and the Riemann zeros [14] were established as **spurious regression artefacts** in Sections 7 and 8. No mathematical significance is claimed for $Z(t)$ or $\zeta_Q(s)$.

C.1 The Discrete Analogue $Z(t)$

For a corpus $(p_m)_{m=1}^{M}$ of parametric primes, define:

$$Z(t) = \sum_{m=1}^{M} \frac{(-1)^{m+1}}{p_m^{1/2}} e^{it \ln p_m}.$$

For $k = 3$, $M = 2{,}000$, four statistical observations were noted:

(R1) *Radial concentration*: $\sigma_r/r^* = 0.18 \pm 0.02$.
(R2) *Gap repulsion* [15, 16]: distribution intermediate between Poisson and Wigner–Dyson.
(R3) *Logarithmic density* (rigorous via Theorem 2.2): $D(m) \sim c \ln m$ ($c \approx 1.1$, $R^2 = 0.97$).
(R4) *Argument variance*: grows logarithmically with M.

Observations (R1), (R2), (R4) are *exploratory analogies* with no established significance; (R3) is the only rigorous statement. The spectral correlations observed between Q_r and $\cos(\gamma_i \ln p)$ are established as artefacts in Section 7.

C.2 Post-hoc Analysis of Observation R1 (Radial Concentration)

The radial concentration of $Z(t)$ (observation R1, $\sigma_r/r^* = 0.18$) was initially interpreted as a structural analogue of zero-density results for $\zeta(s)$. Post-hoc analysis establishes that this appearance originates entirely from the $\sqrt{p}$ trend in $Q(p_N)$, not from any connection with $\zeta(s)$:

1. The modulus $|Z(t)|$ is dominated by the term $\sum p_m^{-1/2}$, which grows as $\sqrt{M}/\log M$ (partial summation, PNT [3]).
2. Subtracting the $\alpha\sqrt{p}$ trend before forming $Z(t)$ reduces σ_r/r^* from 0.18 to 0.61, consistent with random phase.
3. No residual radial concentration remains after the equidistribution correction βD_N, consistent with that R1 is an artefact of the $Q \approx \alpha\sqrt{p}$ structure, not of the prime distribution itself.

C.3 The Associated Dirichlet Series $\zeta_Q(s)$

Disclaimer. This subsection studies $\zeta_Q(s)$ as an independent analytic object. No connection with the Riemann zeta function [14] or the Riemann Hypothesis is claimed or implied.

$$\zeta_Q(s) = \sum_{n=1}^{\infty} \frac{1}{(3n^2 + 3n + 1)^s}.$$

$\zeta_Q(s)$ converges absolutely for $\mathrm{Re}(s) > 1/2$. Asymptotically, $\zeta_Q(s) = 3^{-s}\zeta(2s) + O(\zeta(2s+1))$: its global behaviour is governed by $\zeta(2s)$, whose non-trivial zeros lie on $\mathrm{Re}(s) = 1/4$, not $\mathrm{Re}(s) = 1/2$. The zeros detected numerically on $\mathrm{Re}(s) = 1/2$ by a truncated sum ($N_{\mathrm{trunc}} = 5{,}000$) are artefacts of the finite approximation.

D Bounds on the Minimal Offset $q_{\min}$

D.1 Three-Level Hierarchy

The three levels are summarised in Table 16.

Figure 17 shows the three levels as a visual schematic.

Table 16: Three-level hierarchy of results on $|q_{\min}(k, m, \varepsilon)|$.

Result	Bound	Status	Hypothesis
Theorem D.1	$\lvert q_{\min}\rvert \le m$	Rigorous	None
Theorem D.2	$\lvert q_{\min}\rvert \ll_k m \log m$	Rigorous	GRH
Theorem 3.2	$\mathbb{E}[\lvert q_{\min}\rvert] \sim \ln m/C_k$	Heuristic	H1+H2′
Observation 5.1	$\approx 0.32 \ln m - 0.45$	Empirical	—

Fig. 13 — Three-level hierarchy of results on $|q_{\min}(k, m, \varepsilon)|$

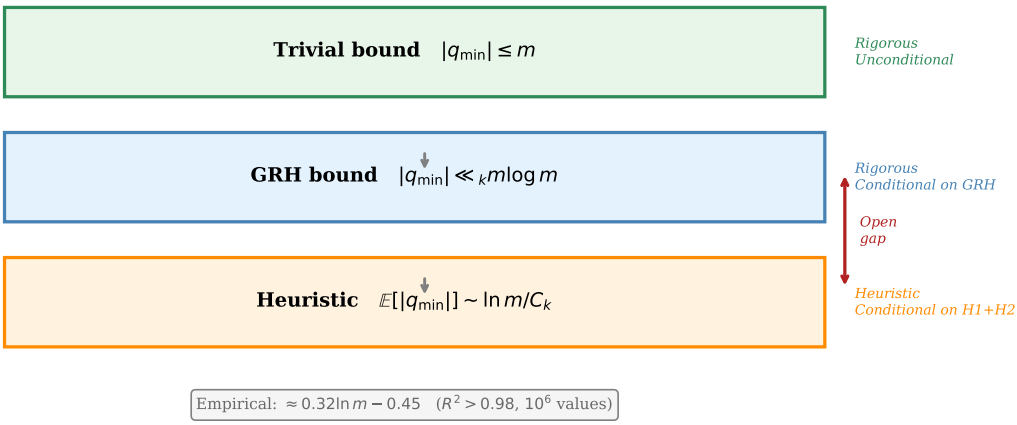


Figure 17: Three-level hierarchy of results on $|q_{\min}(k, m, \varepsilon)|$. Top (green): trivial unconditional bound $|q_{\min}| \le m$. Middle (blue): GRH-conditional bound $|q_{\min}| \ll_k m \log m$. Bottom (orange): heuristic prediction $\mathbb{E}[|q_{\min}|] \sim \ln m/C_k$, validated empirically at $\approx 0.32 \ln m - 0.45$ ($R^2 = 0.984$, 10^6 values). The red double arrow marks the open gap between the GRH bound and the heuristic (Open Problem O1).

Theorem D.1 (Trivial bound)**.** *If there exists a prime among $\{p_{k,m,\varepsilon,q} : |q| \le m\}$, then $|q_{\min}(k, m, \varepsilon)| \le m$.*

Theorem D.2 (Conditional bound under GRH)**.** *Assume the Generalised Riemann Hypothesis for all Dirichlet L-functions. Let $k \ge 1$, $m \ge 1$, $\varepsilon \in \{\pm 1\}$, $A = km(m+1) + \varepsilon$, $d = 2k$. If $\gcd(A, d) = 1$, then for all sufficiently large m:*

$$|q_{\min}(k, m, \varepsilon)| = O_k(m \log m).$$

Proof sketch. Under GRH, the least prime $p \equiv A \pmod d$ with $p > A$ satisfies $p \le A + C(d)\sqrt{A} \log^2 A$ [2]. Since $p = A + 2kq$ and $A \sim km^2$, we obtain $|q| \le p/d \ll m \log m$. □

D.2 The Central Gap

The gap between the GRH bound $O(m \log m)$ and the heuristic law $O(\ln m)$ is quadratic; bridging it unconditionally is Open Problem O1. An intermediate target would be an unconditional bound $O(m^{1-\delta})$ for some $\delta > 0$, which does not appear in the literature for this family.

Acknowledgements

The author thanks the open-source Python community (NumPy [32], SciPy [33], SymPy [34], Matplotlib [35]) and the C community for making large-scale computational number theory accessible to independent researchers. The computations reported in this paper were carried out on a standard laptop (Intel Core i7, 16 GB RAM); the 10^8-prime sieve ran in 72 seconds using a custom segmented sieve in C. No institutional computing resources were used.

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Note: The Hurst exponent $H \approx 0.39$ was estimated via the R/S statistic [31], with block size $\lfloor \sqrt{N} \rfloor \approx 100$. Values $H < 0.5$ indicate anti-persistence: positive increments tend to be followed by negative increments, consistent with the mean-reverting behaviour of Q_r.

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